

A grayscale image of a satellite in space, featuring a large central dish and two long solar panel arrays extending outwards. The background shows a textured surface, likely the Earth's cloud cover.

→ ANNUAL REPORT 2016

ANNUAL REPORT 2016 SCIENCE HUMAN & ROBOTIC EXPLORATION EARTH OBSERVATION TELECOMMUNICATIONS & INTEGRATED APPLICATIONS GALILEO & NAVIGATION-RELATED ACTIVITIES SPACE TRANSPORTATION OPERATIONS TECHNICAL & QUALITY MANAGEMENT HUMAN RESOURCES FACILITY MANAGEMENT & INFORMATICS CORPORATE REFORMS PROCUREMENT, FINANCIAL OPERATIONS & LEGAL AFFAIRS POLICIES PLANNING AND CONTROL DIRECTOR GENERAL'S CABINET

→ ANNUAL REPORT 2016

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ESA Communication Department
ESTEC, PO Box 299
2200 AG Noordwijk
The Netherlands
T +31 71 565 3408

www.esa.int

→ FOREWORD

The world is changing: in addition to fast-moving political developments in both East and West, new technologies and methods of organising projects are either ready for use or in development. Space 4.0 was a major theme of the Council meeting at ministerial level in 2016 and covers a broad variety of aspects including commercialisation, participation, diversity and digitalisation.

These aspects open up fresh opportunities for space in the form of new technologies of which bio-inspired systems, self-healing, additive manufacturing, nano-tubes, in-space production and assembly, artificial intelligence and quantum technology are just a few examples. The latter provides the perfect illustration of the transition from pure academic research to major development activity: for a long time, quantum technology, most notably quantum entanglement, was a subject really only of interest in the laboratories of universities and similar institutions.

However, today, we see quantum technology entering the communication sphere and being harnessed to secure the transmission of key data. This will be of particular significance for the future of autonomous systems, be it in cars, ships, trains or aircraft.

This example of quantum technology provides further compelling evidence for the view that the chain of innovation from pure fundamental research to innovative products in the market requires institutions that take on the role of making the chain seamless. ESA, with its wide range of programmatic activities, is in a position to play such a role and even to broaden this seamless chain of innovation to encompass not just the various space domains but also the non-space sector.

The establishment of new ESA Business Incubation Centres in our Member States as well as the 'Discovery, Preparation and Technology Development' programme are, together with the ESA labs at various universities and the Grand Challenge competition scheme, activities which complement traditional space projects and pave the way for a broad-based, future-oriented ESA.

These various activities require very agile behaviour within the Agency as the 'wind of change', we are accustomed to referring to is not just some weather event to be observed from afar; it also has a real impact on our day-to-day work. In this sense, our internal structures, our interaction with Member States, as well as our relations with industry and academia, have to be adjusted dynamically to address new challenges and ensure that we transform challenge and risk into opportunity.

In this respect, our relationship with the European Commission is one further element serving to cement ESA's role as 'The European Space Agency'. Copernicus, Galileo and EGNOS are all projects that demonstrate the power of close interaction between the EU and ESA – despite our very different structures and rules – as long as the will to cooperate exists on both sides. European citizens and taxpayers deserve an optimised public space sector, and ESA is ready and eager to take its share of responsibility in delivering this.

*Bo Andersen
Jean-Yves Le Gall
Co-Chairs of Council*

→ YEAR IN REVIEW

This year, once again, ESA demonstrated excellence across a range of areas, as outlined hereafter. Following on from past achievements, the Agency obtained the sustained commitment of its Members States at the Council meeting at ministerial level, based on the vision of a 'United Space in Europe' in the Space 4.0 era.

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The LISA Pathfinder mission, launched on 3 December 2015, began its science operations on 1 March 2016, following a very successful payload commissioning. The first mission results appeared on 7 June in the *Journal Physical Review Letters*. The exceptional performance of LISA Pathfinder demonstrated Europe's leadership in the exploration of the 'gravitational Universe'.

After 12 years in space, the Rosetta spacecraft terminated its mission on 30 September 2016. The last images received from the probe show the comet's surface at an unprecedented level of detail, with the very last image having been taken at a distance of only approximately 20 m from the comet's surface, seconds before final touchdown. Now both Rosetta and Philae will ride with the comet for the rest of its orbit, after having changed – with their unique observations – our understanding of these relics of the formation of the Solar System.

The ExoMars 2016 mission was launched from Baikonur on 14 March. The Trace Gas Orbiter and Schiaparelli (an entry-descent-landing demonstrator module) separated on 16 October as planned. The TGO Mars orbit insertion was completed successfully on 19 October and the descent of Schiaparelli towards the Red Planet's surface provided the test data sought for the entry/descent phase even though a final anomaly resulted in a destructive landing.

The Sentinel-3A satellite was put into orbit on a Rockot launcher from Plesetsk, Russia, on 16 February 2016. It is the third Sentinel in orbit, providing the 'bigger picture' for Europe's environmental monitoring programme, Copernicus. The Sentinel-1B satellite was launched on 25 April 2016 on a Soyuz rocket from Europe's Spaceport in Kourou, French Guiana. This second Sentinel-1 satellite provides more 'radar vision' for Copernicus.

The Galileo GNSS constellation continued to be deployed progressively in batches: two satellites were successfully launched by Soyuz (VS15) on 24 May and the first four-satellite launch, on board an Ariane 5 ES launcher adapted to Galileo missions, took place on 17 November.

On 15 January, on the International Space Station, Tim Peake, the British ESA astronaut, performed a 4-hour 43-minute spacewalk together with NASA astronaut Tim Kopra; their six-month mission to maintain the weightless research laboratory and run scientific experiments for hundreds of researchers on Earth ended with a landing in a Soyuz spacecraft on 18 June. The French ESA astronaut Thomas Pesquet, NASA astronaut Peggy Whitson and Roscosmos Commander Oleg Novitsky were launched in their Soyuz MS-03 spacecraft from the Baikonur Cosmodrome in Kazakhstan on 17 November; during the mission, Pesquet took part in many research activities for the Station partners. Their missions were part of ESA's vision to use Earth-orbiting spacecraft as a place to live and work for the benefit of European society while using the experience to prepare for future voyages of exploration further into the Solar System.

Finally, an Association Agreement was signed on 5 July in Paris with the Slovenian Minister of Economic Development & Technology, Zdravko Počivalšek, thus allowing direct participation in ESA optional programmes and the Slovenian authorities to be represented in the relevant delegate bodies. The Agreement entered into force on 16 November.

Jan Wörner
Director General

→ DIRECTORS



Jan Wörner
Director General



Alvaro Giménez Cañete
Directorate of Science



Josef Aschbacher
Directorate of Earth
Observation



Franco Ongaro
Directorate of Technology,
Engineering and quality



David Parker
Directorate of Human and
Robotic Exploration



Paul Varhoef
Directorate of Galileo Programme
& Navigation-related Activities



Rolf Densing
Directorate of Operations



Daniel Neuenschwander
Directorate of Launchers



Magali Vaissiere
Directorate of Telecommunications
& Integrated Applications



Eric Morel de Westgaver
Director of Industry, Procurement
and Legal Services



Jean Max Puech
Directorate of Internal Services

→ OUTCOME OF THE 2016 MINISTERIAL COUNCIL

On 2 December 2016, ESA concluded the two-day Ministerial Council meeting in Lucerne, Switzerland. Ministers in charge of space matters from ESA's 22 Member States plus Slovenia and Canada allocated €10.3 billion for space activities and programmes based on the vision of a United Space in Europe in the era of Space 4.0.

Following on from past achievements, the Agency obtained the sustained commitment of its Member States at the Council meeting at ministerial level, based on the vision of a 'United Space in Europe' in the Space 4.0 era.

The high level of subscriptions demonstrates once more that ESA's Member States consider space to be a strategic and attractive investment having particularly high socio-economic value. It also underlines that ESA is truly the European space agency capable of channelling their investment so as to respond effectively to regional, national and European needs by covering all areas of space activity: science, human spaceflight, exploration, launchers, telecommunications, navigation, Earth observation, applications (combining space, airborne and terrestrial technology), operations and technologies; as well as responding to the needs and challenges of Europe and the Member States by bringing together all stakeholders.

The Ministers confirmed once again their confidence that ESA is capable of conceptualising, shaping and organising change in the European space sector and at ESA itself. While also acting as a global player, broker and mediator in the very centre of international cooperation in space activities, in areas ranging from the far away in exploration (with the 'Moon Village' concept, for instance) to supporting, closer to home, the international global climate research effort following the Paris Agreement of 2015.

At this meeting, the Ministers declared their support for the ESA Director General's vision for Europe in space and the role and development of ESA: now the Space 4.0 era can start with ESA committing to inform, innovate, interact and inspire while also building on commercialisation, participation, digitalisation, jobs and growth to allow the 'United Space in Europe' concept to continue to unfold.

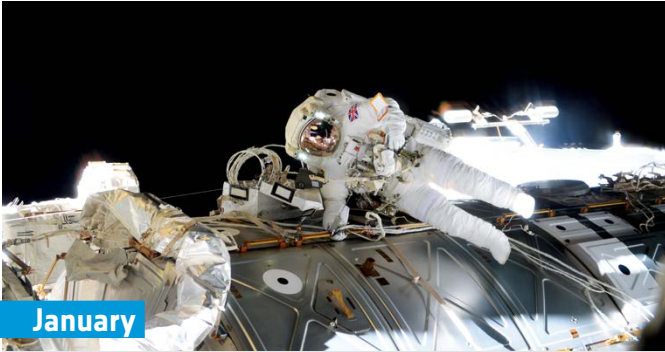
The following Resolutions were adopted at the 2016 Ministerial Council:

- Towards Space 4.0 for a United Space in Europe
- Level of Resources for the Agency's Mandatory Activities 2017–21
- Guiana Space Centre (CSG), 2017–21
- ESA Programmes



European ministers and representatives at the ESA Ministerial Council in Lucerne, on 1-2 December 2016

→ HIGHLIGHTS OF 2016



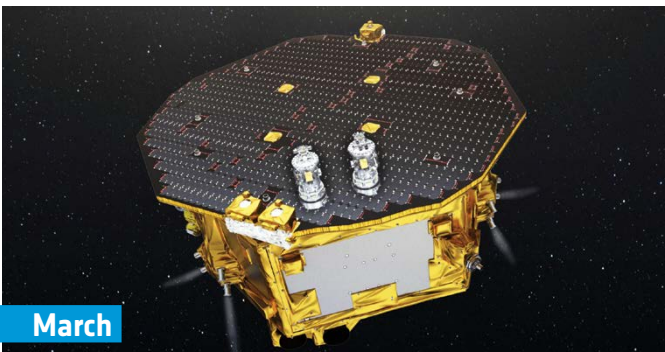
January

15 – ESA astronaut Tim Peake performs a 4-hour 43-minute spacewalk from the Space Station



February

16 – Sentinel-3A launched on a Rockot launcher from Plesetsk, Russia



March

3 – LISA Pathfinder starts science operations following launch on 3 December 2015
14 – ExoMars 2016 launched from Baikonur, Kazakhstan



April

25 – Sentinel-1B launched from Europe's Spaceport, French Guiana



May

24 – Two Galileo GNSS constellation satellites launched on Soyuz VS15 from Europe's Spaceport



June

1 – First Sentinel-1 images delivered after being sent via the European Data Relay System's laser technology
18 – ESA astronaut Tim Peake returns back to Earth in Soyuz TMA-19M spacecraft



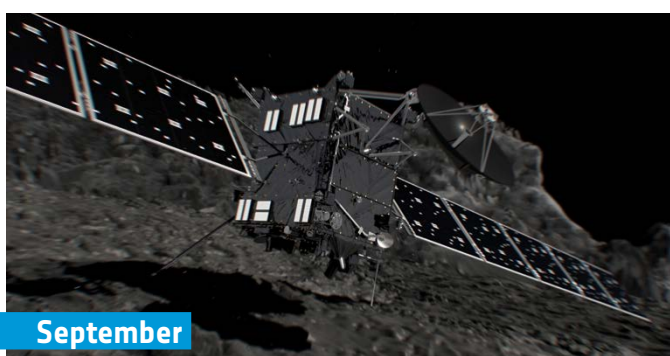
July

5 – ESA Director General Jan Wörner and Slovenian Minister of Economic Development Zdravko Počivalšek sign an Association Agreement



August

30 – First Ariane 5 rocket modified to carry four Galileo satellites arrives at Europe's Spaceport



September

30 – Rosetta mission ends



October

19 – ExoMars Trace Gas Orbiter enters Mars orbit



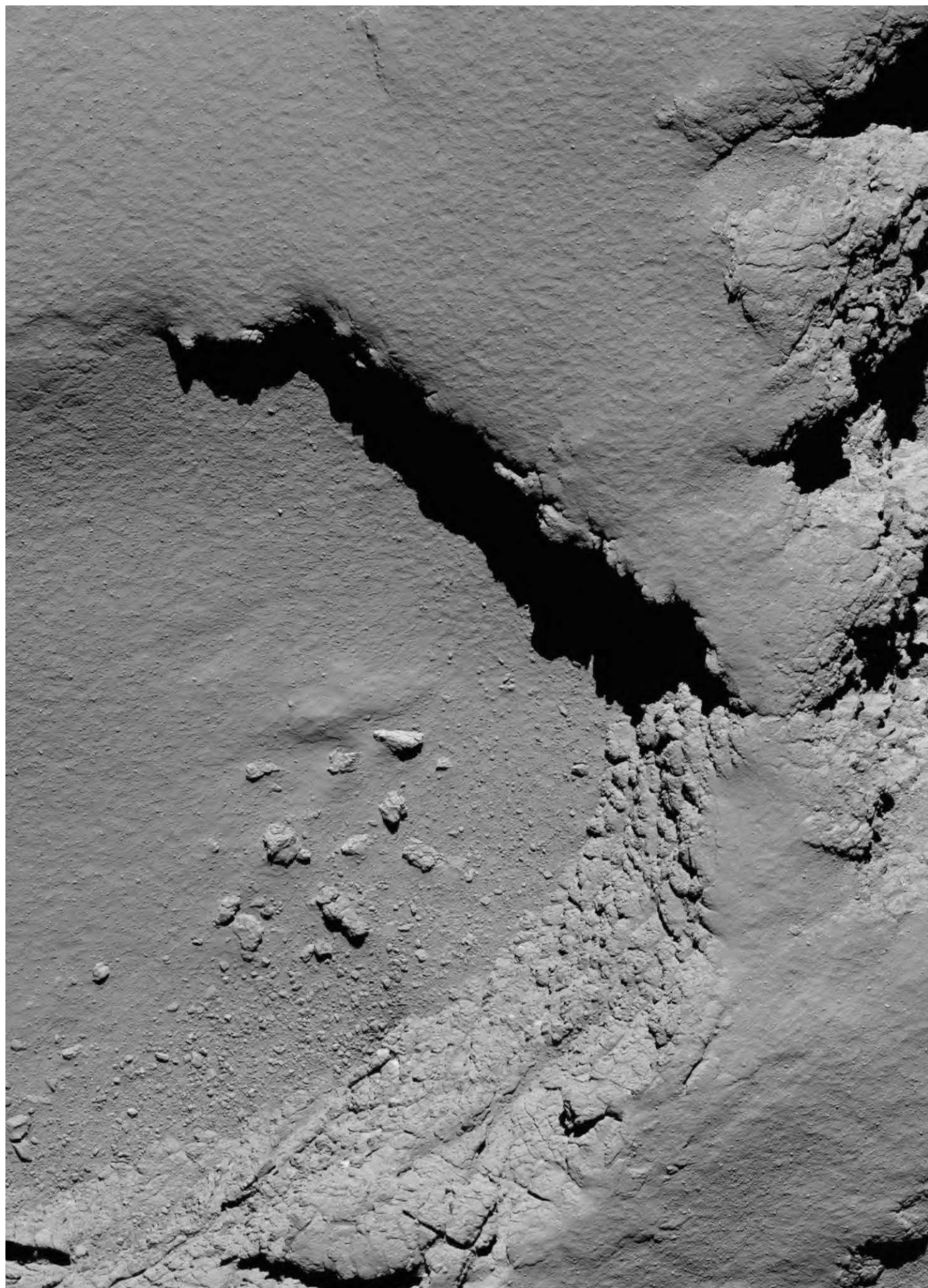
November

17 – First four-satellite Galileo launch on adapted Ariane 5 ES
17 – ESA astronaut Thomas Pesquet launched on six-month Proxima mission



December

2 – Ministerial Council 2016 in Lucerne, Switzerland



Cosmic Vision

Science Programme planning progressed throughout the year, with replanning of activities discussed with the Science Programme Committee to ensure a vibrant programme with content compatible with the decisions taken at Ministerial Council 2016.

The definition phase of the M3 mission candidate Plato, and the initial study phase of three M4 candidates were completed. The M4 candidates are Ariel, investigating the atmospheres of exoplanets; Thor, investigating the energy balance and particle acceleration in the solar wind; XIPE, performing X-ray polarimetry from a variety of astrophysical objects.

A total of 25 proposals was received in response to the call for the M5 mission issued in April, while one proposal was received in response to the call for the L3 mission, addressing the science theme 'The Gravitational Universe', to be pursued by implementing a gravitational wave observatory with a planned launch date of 2034.

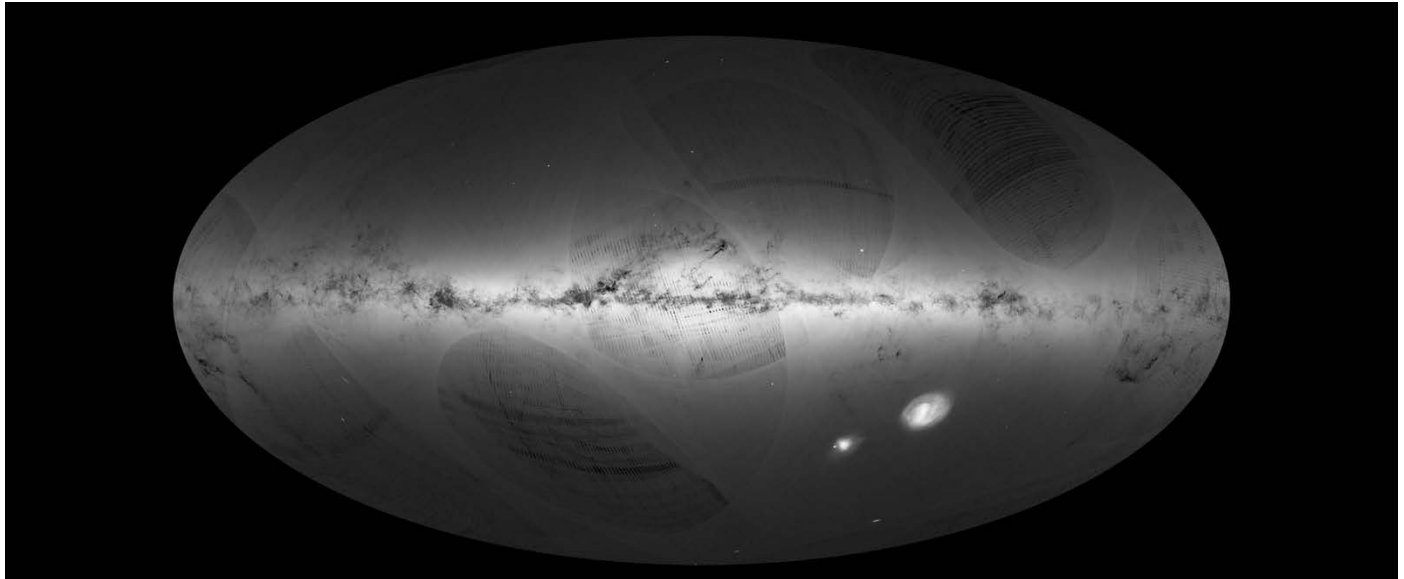
Missions in operation

Missions in operation at the end of 2016 included astrophysical observatories and surveyors (Gaia; Integral; XMM-Newton; and Hubble Space Telescope with NASA), in situ missions to planets and minor bodies in the Solar System (Mars Express; Cassini with NASA), spacecraft studying the Sun (Proba-2, SOHO and Iris with NASA, Hinode with JAXA) and the Sun-Earth connection (Cluster) and LISA Pathfinder, demonstrating technology for the detection of gravitational waves. Post-operational data calibration and processing activities were continuing for Herschel, Planck and Rosetta, and were completed for Venus Express.

The Rosetta spacecraft continued to accompany Comet 67P/Churyumov-Gerasimenko on its journey around the Sun. In September, following a controlled approach, the spacecraft landed on the comet in a high-profile end-of-mission finale. Unique scientific results on the physical, chemical, and electrical properties of the comet were returned during this very close approach and landing phase. Shortly before the Rosetta end-of-mission, the Philae lander was discovered on the surface of the comet. The mission focus switched to ensuring that its scientific legacy is captured by beginning the processing and ingesting science-ready data into European Science Data Centre archives.

Gaia passed its first major scientific milestone in September with the first Gaia Data Release (DR1) presenting the positions

Comet 67P/Churyumov-Gerasimenko seen during the Rosetta spacecraft's final descent on 30 September (ESA/Rosetta/MPS for OSIRIS Team)



An all-sky view of stars in our galaxy – the Milky Way – and neighbouring galaxies, based on the first year of observations from ESA's Gaia satellite, from July 2014 to September 2015

and respective brightness of approximately one billion stars. DR1 also included the Tycho Gaia Astrometric Solution catalogue with distances and proper motions for a subset of two million bright stars in common with the Hipparcos mission. The release of DR1 has led to many science papers covering topics ranging from the distance scale of the Universe, stellar evolution, the dynamics of galaxies, and the discovery of exoplanets.

LISA Pathfinder was commissioned, and completed ground-breaking low-noise drag-free measurements during the year that far surpassed expectations, despite a number of technical anomalies. The results confirm ESA's leadership in the race to operate a space-based gravitational wave observatory.

XMM-Newton continued to deliver a steady flow of papers and an update of the serendipitous source catalogue was published. Integral has found an interesting new role as the most sensitive all-sky detector for potential electromagnetic counterparts to gravitational wave and astrophysical neutrino events.

Science archives

The ESA science archives, together with the newly developed ESASky portal, provide data from more than 15 current and past space science missions, and are fundamental tools used by thousands of space scientists worldwide. These web applications allow scientists openly and seamlessly to select, compare, and retrieve science-ready data and catalogues from space missions in astronomy, Solar System science, and heliophysics.

Research activities

Research areas in 2016 included Earth's magnetosphere, meteor showers, heliophysics, the terrestrial planets, the formation and evolution of moons, planets, stars, and galaxies, fundamental physics and cosmology.

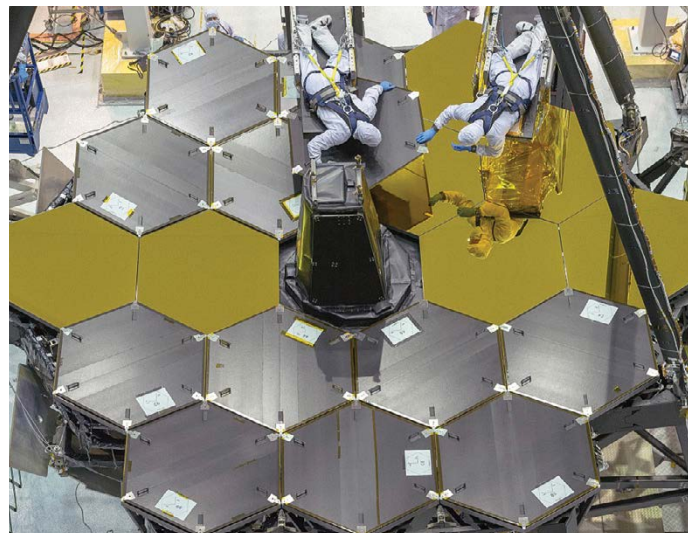
One project involves the digitisation and analysis of Carte du Ciel photographic sky plates obtained from 1887 onwards at a number of observatories. Comparison of astrometry and photometry with recent Gaia measurements allows studies of binary stars over periods from several decades to about a century; variability of stars on a century timescale; and orbital variations of asteroids.

Projects under definition & development

BepiColombo

Launch: April 2018

BepiColombo will make a detailed study of Mercury, the least explored planet in the inner Solar System. The acceptance



Inside the cleanroom at NASA Goddard on 25 April, the James Webb Space Telescope's primary mirrors are uncovered in preparation for instrument installation (NASA/C. Gunn)

testing of the flight models of both ESA-provided modules, the Mercury Planetary Orbiter and the Mercury Transfer Module, as well as the JAXA-provided Mercury Magnetospheric Orbiter, are continuing.

James Webb Space Telescope

Launch: 2018

The James Webb Space Telescope is a 6.5-m infrared observatory, under joint development by NASA, ESA and the Canadian Space Agency. It will search for light from the first objects that formed in the Universe after the Big Bang, study galaxy evolution, and probe the early phases of star and planet formation. The two European instruments (NIRSpec and MIRI) are integrated into the payload module and are participating in the flight model environmental test campaign.

CHEOPS

Launch: 2018

CHEOPS is a small spacecraft devoted to characterising exoplanets orbiting bright stars via the transit method. Development is on track for a launch at the end of 2018 (or later) as a passenger with a larger satellite.

Solar Orbiter

Launch: 2019

Solar Orbiter will provide unprecedented close-up and high-latitude observations of the Sun and inner heliosphere. The flight models of the scientific instruments are being integrated into the spacecraft and the flight model acceptance test campaign is about to begin.

Euclid

Launch: 2020

Euclid will investigate the nature and origin of dark matter and dark energy. The system design is progressing well and the Critical Design Review process will start at the end of 2017. Progress on the development models (structure and thermal, as well as engineering models) is mostly nominal. Some delays have been experienced in the flight model development of the payload module, the Visible Instrument, and the detectors for

the Near Infrared Instrument, and are being addressed through dedicated action plans.

JUICE

Launch: 2022

The JUICE mission will explore the Jupiter system and its icy satellites, with an emphasis on Ganymede, Callisto, and Europa. After a two-year tour of the Jupiter system, the spacecraft will enter orbit around Ganymede. The spacecraft Preliminary Design Review is expected to be completed in March 2017 after which the detailed design phase will get under way.

PRODEX

PRODEX (Programme de Développement d'Expériences Scientifiques) manages around 200 contracts, and in 2016 started or continued instrument developments for ESA science, Earth observation, and exploration. Three new Participating States joined PRODEX at Ministerial Council 2016, bringing the total to 14.

After the launch in March of ExoMars 2016, the NOMAD (atmospheric spectroscopy – BE) and CaSSIS (colour stereo imager – CH, PL, IT) instruments were commissioned during the cruise phase.

Phase-C/D development of Solar Orbiter instruments continued for EU1 (BE) and STIX (CH), and for subsystems of RPW, SPICE, METIS and PAS-SWA, representing 20 contracts in six countries. Flight model deliveries are scheduled for May–July 2017.

Significant progress has been made on Phase-B2 of the JUICE instrument subsystems (10 activities in AT, BE, CZ, PL and CH contributing to seven instruments).

After preparatory activities in PRODEX and System Requirements Review completion, the ALTIUS small mission (BE) for the monitoring of stratospheric ozone has been transferred to ESA's Earth Observation Programmes (EOP) and proposed as an element of the European Earth Watch Programme. Further support will be given to ALTIUS in the next phase via PRODEX in cooperation with EOP.





human & robotic exploration

Highlights

At the European Astronaut Centre in May, the ESA Director General announced in the presence of German Chancellor Angela Merkel that Alexander Gerst had been assigned to a second long-duration mission to the International Space Station (ISS), to take place in 2018. During the second half of the mission, Alex will be ISS Commander.

Tim Peake, NASA astronaut Tim Kopra and Roscosmos Commander Yuri Malenchenko landed on 18 June in their Soyuz TMA-19M spacecraft. Tim was the eighth ESA astronaut to complete a long-duration mission in space. On 17 November, Thomas Pesquet was launched on Soyuz MS-03 from Baikonur for a long-duration mission.

The ExoMars Trace Gas Orbiter (TGO) and Schiaparelli demonstrator were launched together on 14 March on a Proton rocket from Baikonur. The two separated on 16 October as planned. While the TGO Mars orbit insertion was completed on 19 October, an anomaly occurred during the descent of Schiaparelli towards the surface of Mars, resulting in a destructive landing. The Schiaparelli Anomaly Investigation Team (SAIT) was set up immediately to determine the cause of the failure. In parallel the ESA Inspector General was tasked with setting up an independent Schiaparelli Investigation Board composed of external experts including NASA JPL and industry, to verify/confirm the findings of the SAIT.

At the Ministerial Council 2016, ESA Member States took the strategic decision to gather in one new programme, the European Exploration Envelope Programme (E3P), all activities relating to human and robotic exploration. The aim of this programme is to implement the ESA Space Exploration Strategic Vision of bringing the benefits of exploration to society in the areas of science, economics, inspiration, and global cooperation.

Member States also decided to continue to operate and use the ISS until 2024 together with their international partners and they provided funding for ExoMars 2020, for the European contribution to the Russian-led Luna-Resurs Lander, for studying future missions and to start the development of the related enabling technologies.

ESA's European Service Module arrived at NASA Glenn's Plum Brook Station for testing (NASA)



The ExoMars 2016 mission was launched on 14 March on a Proton rocket from Baikonur

Research

Astronauts

On 15 January 2016, Tim Peake and Tim Kopra performed a 4-hour 43-minute spacewalk to replace a failed power regulator and to install cabling. The sortie was stopped early after Tim Kopra reported a small amount of water building up in his helmet. After completion of their main task, they were instructed to return to the airlock.

A team from EAC implemented the Cooperative Adventure for Valuing and Exercising human behaviour and performance Skills (CAVES) 2016 course in Sardinia (IT) in July. CAVES 2016 was the first training course where astronauts from NASA, JAXA, Roscosmos and ESA (Pedro Duque) trained with a Chinese taikonaut.

European Robotic Arm (ERA)

The Multipurpose Laboratory Module (MLM) experienced propulsion system contamination issues. As a result, the MLM and ERA launch will be further delayed –it is predicted until the second half of 2018. Plans were made for moving ERA to Baikonur and starting cosmonaut training.

Atomic Clock Ensemble in Space (ACES) and Atmosphere–Space Interaction Monitor (ASIM)

Both the Atomic Clock Ensemble in Space (ACES) and the Atmosphere–Space Interaction Monitor (ASIM) were confronted with technical issues causing delays, extra costs and the need for constant monitoring by ESA. Good progress was made on solving the issues. The launch of ACES was delayed until spring 2018 while the ASIM launch date is scheduled for the end of 2017, also depending on SpaceX and the ISS manifest.



ESA astronaut Tim Peake during his spacewalk on 15 January (ESA/NASA)

Research

European Programme for Life and Physical Sciences and Applications in Space (ELIPS)

Science roadmaps covering a broad spectrum of disciplines and providing a strategic framework for human spaceflight research activities in the coming years were defined with the science community. Each roadmap helps select and focus individual projects towards a set of overarching goals. They address objectives that will help to enable and expand human presence in space and address global development goals on Earth.

ISS Utilisation

In life sciences, 11 different astronauts were involved in ESA activities on the ISS as test subjects in 2016. The research focused on improved understanding of the effect of long-duration spaceflight on: alterations in circadian rhythms; existence, extent and mechanisms of post-reentry bone loss; complex interactions of cognition, stress and immunity; pulmonary function; headache incidence; and energy balance.

In the physical sciences, two sets of alloy samples completed processing in the Electromagnetic Levitator while the Batch 2b alloy solidification experiments started processing in the Materials Science Laboratory.

In complex plasma physics, the joint ESA/Roscosmos Plasma Kristall-4 facility completed two science campaigns. The SODI DCMIX-3 experiment was completed. A new personal active astronaut radiation dosimetry system began testing while the DOSIS-3D payload continued to monitor the radiation environment in Columbus through the year, and results were published.

Definition work continued on 22 biology and exobiology and 11 human research proposals for future ISS experiments from the International Life Science Research Announcement (ILSRA) 2014.

Hardware and missions of non-ISS platforms

The Chinese SJ-10 orbital capsule was launched in April. It included ESA's Soret Coefficient in Crude Oil experiment as ESA's first ever payload to be carried on a Chinese space mission.

The 64th and 65th ESA Parabolic Flight Campaigns were performed with 23 ESA experiments across the two campaigns including the validation of the Vibration Induced Phenomena in GRANular media (VIP-GRAN) design concept for ISS, and four student experiments from the 'Fly Your Thesis!' programme.

A dedicated Announcement of Opportunity was released for an International Space Life Science Working Group Parabolic Flight Campaign, which will include lunar/martian gravity levels.

Twelve drop tower campaigns were completed in 2016 covering a wide range of science and technology objectives, including support to one of the MAXUS-9 sounding rocket experiments and to perform a student 'Drop Your Thesis' campaign.

The 60-day 'Reactive Jump' bedrest study was completed at the DLR:envihab facility in Cologne, Germany. A Concordia Antarctic winter isolation campaign was completed including five ESA experiments.

Exploration

Multi-Purpose Crew Vehicle/European Service Module (MPCV/ESM)

The European Service Module Test Article (E-STA) was fully assembled and mated in Plum Brook (US) on schedule. All tests were performed, the E-STA was accepted and ownership transferred to NASA.

The Critical Design Review closeout board was held on 31 October. A number of major critical issues remained outstanding so that the ESM1 delivery on 30 April 2017 was no longer possible. The expected slippage of the delivery date was not accepted by ESA/NASA, and a schedule acceleration plan was delivered by Airbus on 15 December.

MPCV-ESM2

The ESM2 price and associated conditions were negotiated and agreed with Airbus. The formal contract update was targeted for November, with formal signature expected to occur only after Ministerial Council 2016. The ESM2 barter value was agreed with NASA.

Multi-Purpose End-To-End Robotic Operation Network (METERON)

Tim Peake performed the Supvis-M (supervisory control from ISS) experiment, piloting the rover Bridget in the Mars yard in Stevenage (UK) from the ISS, at the end of April. Supvis-M is part of the METERON series of experiments investigating various robotics and operations strategies for future exploration missions with human/robotic elements.

International Berthing Docking Mechanism (IBDM)

The Sierra Nevada Corporation (SNC) negotiated with NASA one flight using docking for attaching the Dream Chaser space vehicle to the ISS. A cooperation scheme was developed among ESA, European industry, SNC and the ESA Participating States. In addition, JAXA (for HTV-X) and Lockheed Martin (for Orion) have also expressed interest in IBDM.

Robotic exploration

ExoMars

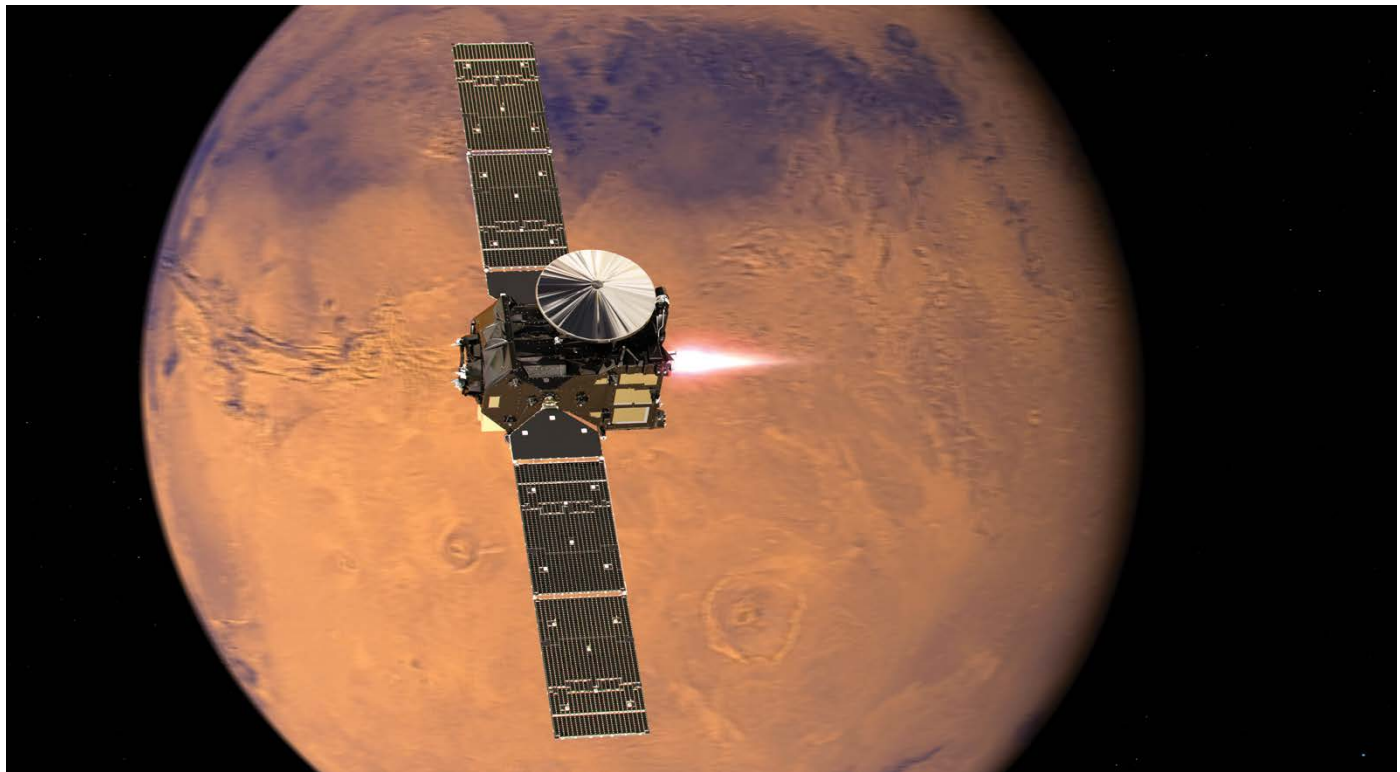
ExoMars 2016 mission

The Schiaparelli mission demonstrated good functionality of all but one subsystem. The anomaly leading to the hard landing was clearly identified, and will provide valuable input for the design and validation of the 2020 entry, descent and landing functionality.

In November, there was a first opportunity to perform some TGO science measurements, primarily for the purposes of calibrating the science instruments.

ExoMars 2020 mission

The ExoMars Lessons Learned Exercise (ELLE) was performed by the ESA Inspector General. The ExoMars project organised periodic information sessions addressing the most critical project domains with the independent ExoMars Tracking Committee (ETC).



Exomars Trace Gas Orbiter enters Mars orbit

The ETC was kept as a permanent independent committee reporting on a regular basis to the HME Programme Board.

In preparation for the Ministerial Council, the Inspector General led the Key Point Review (KPR) with the objective of confirming that technical status, project schedule and financial situation are compatible with a launch date in 2020. Though the results of the review were positive, the Ministerial Council agreed that budget approval is conditional upon a successful Check Point Review in April 2017. During the KPR, a consolidated schedule was drawn up which was assessed as credible by the independent review team.

ExoMars Rider 2 was signed on 16 December consolidating the industrial contract for the full development of the ExoMars Rover and Surface Platform mission in 2020.

European contributions to the Luna-Resurs lander mission

The Joint Work Plan for Lunar exploration in cooperation with Russia was signed by ESA and Roscosmos on 18 May at ESTEC. It covers the initial phase of the cooperation and includes the full development of Platform for Resource Observation and in-Situ Prospecting for Exploration, Commercial exploitation and Transportation (PROSPECT) and the design of Precise Intelligent Landing using Onboard Technologies (PILOT).

The corresponding Phase-B+ activities for PROSPECT and PILOT started in spring, with a contract signed with Leonardo at Farnborough (UK) for PROSPECT, and one with Airbus Defence & Space Bremen at ILA (DE) for PILOT. The first models for the demonstrator of the PILOT Landing Camera ('PILOT-D') on board the Luna-Glob mission were shipped to IKI at the end of 2016.

Exploration Preparation and Technology Development

The assessment study of the Lunar Polar Sample Return mission, together with the Lunar Polar Volatile Prospector rover (two parallel contracts with TAS-I and Airbus Defence & Space) were completed during the summer.

The Phobos Sample Return Phase-A study, funded by the General Studies Programme, was completed in June, indicating feasibility of a mission to be launched in 2025/26. Due to the very high estimated cost-at-completion of the ESA standalone mission option, a small contract was placed in October to investigate a design-to-cost.

Mars Robotic Exploration Preparation (MREP)

The Concurrent Design Facility study on the Mars Sample Return (MSR) carrier was completed with a final presentation on 5 July. ESA and NASA exchanged letters on potential areas for cooperation on Mars exploration.

MREP continues to implement technology activities and studies in support of preparation towards the long-term goal of European participation in an international MSR mission. More than 30 technology activities funded by MREP are currently in implementation.

Commercial partnerships

Eight ideas for partnerships with private sector entities were selected to enter a pilot phase after the 'Space Exploration as Driver for Growth' Call for Ideas. The first four Memoranda of Understanding for the pilot phase of commercial partnership projects were signed. Two projects, ICECubes and Bartolomeo passed technical review.

Strategy

International Cooperation

In preparation for a potential future collaboration with the Chinese Manned Space Agency on the Chinese Space Station (CSS) the three working groups (Utilisation, Astronauts and Infrastructure) have started discussions on areas of common interest to establish long-term cooperation. A meeting was held between ESA and the China National Space Administration on lunar exploration in November in Beijing to review opportunities for cooperation.

International Space Exploration Coordination Group (ISECG)

Discussions have been focusing on agencies' plans for conceptual studies for lunar surface missions and next steps in coordinating agencies' efforts related to lunar polar volatiles characterisation as well as the process for developing the third iteration of the Global Space Exploration Roadmap (GER) for publication at the end of 2017. Common studies with ISECG members to realise the GER continued with the definition of concepts for a human lunar surface demonstrator mission to be implemented in the next decade.

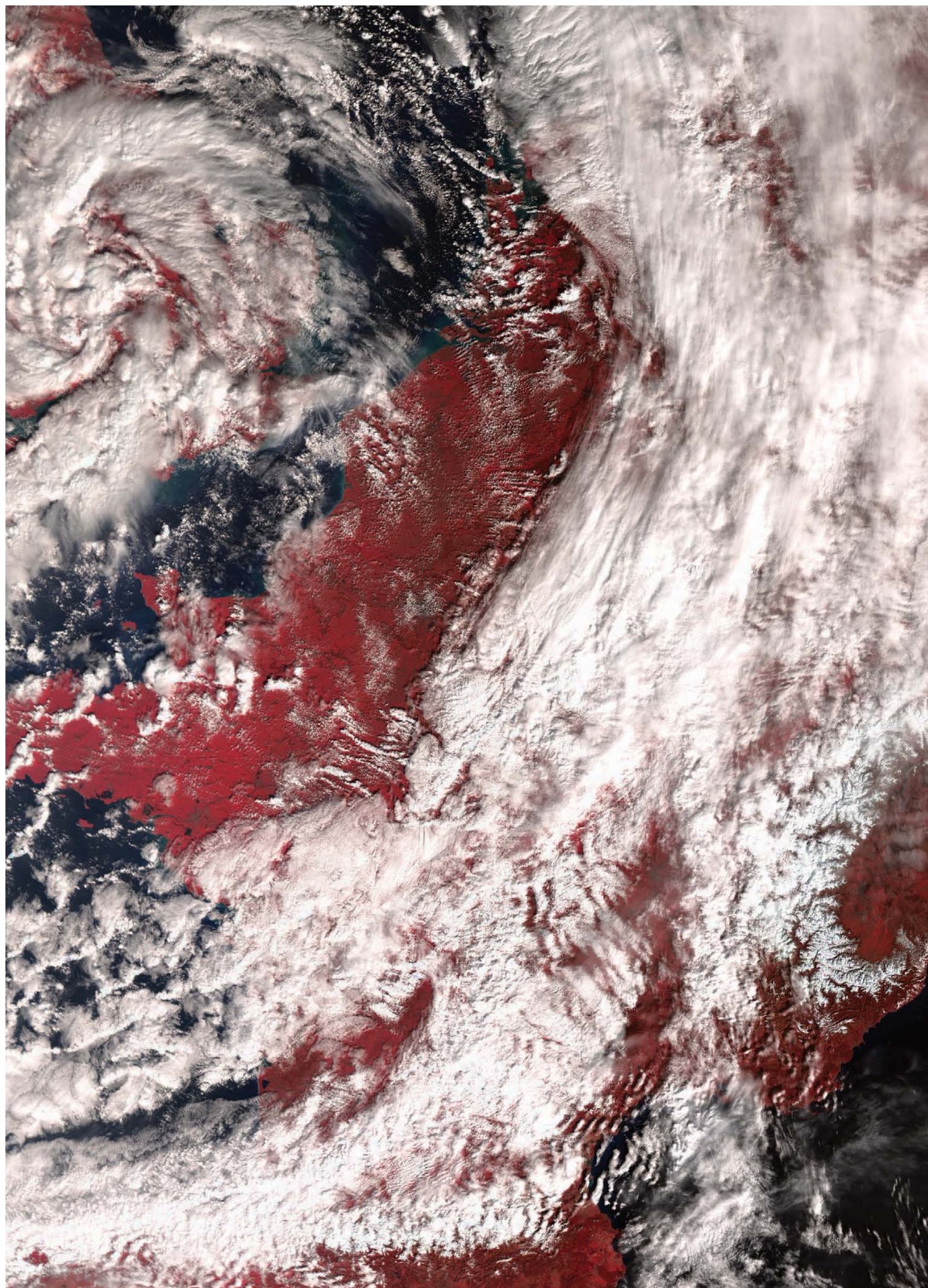
Outreach

The Space4Inspiration event was held in the London Science Museum on 14–15 September. More than 250 participants from all over the world, from commercial and non-space companies, academia, space industry and international organisations attended.

A number of videos were produced which attracted a large viewing on the ESA web site and on YouTube:

- Destination LEO (35,170 views)
- Universe to Explore (31 238 views)
- On the Shoulders of Giants (43 852 views)
- Principia mission videos focused on ISS science (> 1 million views combined across ESA platforms and Tim Peake's Facebook page)

About 8000 people visited EAC in 2016. EAC remains a major outreach factor for human spaceflight activities.





earth observation

Copernicus

Copernicus will provide accurate, timely, easily accessible information to improve management of the environment, understand and mitigate the effects of climate change and ensure civil security.

Recognising a rapidly evolving policy context for the European Union (EU), the 2016 EU Space Strategy introduced new priorities for additional support to EU policies, building upon the initial definition of the scope of the Copernicus Programme. Additional services are being considered to meet emerging needs in priority areas. This 'Copernicus Evolution' will take account of new technical capabilities in the sector, the availability of new data sources and the long-term capacity of the private sector to deliver appropriate solutions.

The European Commission (EC) and ESA carried out consultations with users and Member States to start to develop a Copernicus Space Component Evolution Plan to reflect users' needs, and that ensures coordination with national activities. In response to the priority identified for a CO₂ anthropogenic emission monitoring system in support of the 2015 UNFCCC Paris Climate Agreement, ESA and the EC first responded by jointly establishing a CO₂ Monitoring Task Force. Building on this first Task Force, the EC and ESA established plans to convene additional themed task forces of experts to serve other high-priority needs in support of EU policy domains.

Sentinel-1

Sentinel-1 is the Copernicus high-resolution radar mission. It is based on a constellation of two satellites, orbiting 180 degrees apart.

Sentinel-1A, the first spacecraft in the Sentinel-1 mission has been in operation since October 2014. The second spacecraft, Sentinel-1B, was launched on 25 April and joined operations in September after a detailed phasing and in-orbit calibration campaign, completing the first Copernicus mission in operation.

The procurement of the recurrent Sentinel-1C and Sentinel-1D spacecraft was initiated in December 2015. These new spacecraft will carry a new payload to augment the radar data with Automatic Identification System data to better support Sentinel-1 maritime applications.

Sentinel-2

Sentinel-2 is the optical multispectral Copernicus mission. The mission is based on two satellites flying in the same Sun-synchronous orbit with a separation of 180 degrees.

Sentinel-2A is delivering excellent mission performance. In 2016, Sentinel-2B's integration and test programme was carried

One of the first images from the Sentinel-3A satellite taken on 2 March (Copernicus Sentinel data 2016)



Sentinel-1B lifts off on Soyuz flight VS14 from Europe's Spaceport in French Guiana on 25 April

out, leading to completion of the flight acceptance review in November. Due to the lack of launch opportunities, the launcher was changed from Rockot to Vega for a launch to take place in the first quarter of 2017.

The contract for the Sentinel-2C and Sentinel-2D satellites was signed at the beginning of 2016 and is proceeding.

Sentinel-3

Sentinel-3 is the Copernicus mission delivering global, frequent and near-real time ocean and land monitoring. Two satellites flying in the same Sun-synchronous orbit with a separation of 180 degrees provide a one-day (land) and two-day (ocean) revisit time.

Sentinel-3A was launched on 16 February and is delivering excellent mission performance. Access to the core products was opened for all users starting gradually from October. The second spacecraft, Sentinel-3B, is undergoing final testing. The Sentinel-3B flight acceptance review is scheduled for completion in the final quarter of 2017, with launch date depending on launcher availability. The procurement of the recurrent Sentinel-3C and Sentinel-3D satellite models was initiated in December 2015 and is proceeding.

Sentinel-4

Sentinel-4 is the Copernicus mission that will deliver accurate, hourly data supporting air quality monitoring and forecasting over Europe. Two identical Sentinel-4 instruments will be carried on two corresponding MTG-Sounder (MTG-S) satellites. Phase-C, aimed at detailing the instrument design, was completed with the system Critical Design Review in November.

Sentinel-5

Sentinel-5 is the Copernicus mission providing global data on the composition of the atmosphere. Three identical Sentinel-5 instruments will be carried on the corresponding MetOp-SG satellite A- series.

The Phase-C/D/E1 industrial contract with prime contractor Airbus Defence & Space (DE), which also includes the two recurrent flight models (FM2 and FM3) was signed in June. All subcontractors have been selected and activities begun.

Sentinel-5P

The Sentinel-5 Precursor mission, a single-satellite mission within Copernicus, provides continuity of the atmosphere chemistry mission on Envisat in the period up to the entry into service of the Sentinel-5 operational instrument. It is implemented in cooperation with the Netherlands.

Following the qualification and acceptance review, the satellite and its ground segment are ready awaiting provision of a Rockot launcher for launch by the end of 2017.

Sentinel-6/Jason-CS

The Sentinel-6/Jason-CS mission continues on from the high-precision measurement of ocean topography started with TOPEX/Poseidon in 1992 and carried forward with Jason-1, Jason-2 and Jason-3, and serves as reference for other ocean altimetry missions. There will be two Sentinel-6 satellites in sequence.

The programmatic framework with the international partners was established with the approval of the ESA-NASA-Eumetsat-NOAA Memorandum of Understanding. Flight equipment manufacturing is progressing and the satellite assembly, integration and test activities will begin in summer 2017.



Sentinel-3B's Sea and Land Surface Temperature Radiometer under test at the UK's Rutherford Appleton Laboratory on 1 December

Earth Explorer and Earth Watch missions launched

The Earth Explorer and Earth Watch missions in orbit (SMOS, CryoSat, Swarm and Proba-V) all performed extremely well in 2016. These pioneering missions feature strong elements of international collaboration. In particular, CryoSat and SMOS observed a sharp decline in the polar ice volume and ice extent in the 2016–17 winter. Swarm discovered a 'jet stream' in Earth's core beneath Alaska, Canada and Siberia that is so strong that it also implies that the inner core may be subject to rotation.

Envisat and European Remote Sensing (ERS) activities are in the final phase of the restructuring plan initiated in 2013 with the aim of preparing for the transfer of ERS/Envisat data management to the ESA Heritage Data Programme (LTDP+) during 2017. Access to the data from many Third-Party Missions continued to be provided through the Earthnet programme.

The COSPAR-41 Report comprising input on science activities by ESA, the latter part being mainly dedicated to Earth Explorer Missions, was coordinated by EOP and published in June.

Future Earth Explorer missions

Aeolus

Aeolus will provide the first ever measurements of vertical wind profiles from space. These data are expected to advance numerical weather prediction substantially. There was significant progress related to completion of the instrument, Aladin, and the follow-on completion of the entire satellite, which will undergo final testing. The life test of the laser transmit optics within the instrument was also completed. The entire system including the Flight Operations Segment and the Payload Data Ground Segment are getting ready for the target launch slot in January 2018.

EarthCARE

EarthCARE will carry a payload of four instruments that will operate together with the aim of providing a better understanding of clouds and aerosols and their impact on Earth's climate. It is a joint mission with JAXA (Japan). Production and integration of the three optical instruments has progressed well in Europe. In particular, the final assembly and initial tests of the Protoflight Flight Model (PFM) transmitter of the complex UV lidar instrument. In Japan, the integration of the PFM Cloud Profiling Radar has been completed and JAXA began initial tests of the assembled instrument.

Biomass

The Biomass mission will provide measurements of forest biomass and forest height to advance our understanding of the carbon cycle. The mission will be implemented through a full polarimetric P-band synthetic aperture radar flown on a single satellite. The contract to build the Biomass satellite was awarded to a consortium led by Airbus Defence & Space (UK) and work was initiated in March. The System Requirements Review was completed.

FLEX

FLEX will provide the first global mapping of vegetation fluorescence, to quantify photosynthetic activity and plant stress. A Phase-B2/C/D/E1 contract for the development of the instrument, Floris, was placed with Leonardo (IT). The overall mission, including the resulting data products and the tight link with the convoy companion satellite Sentinel-3, were thoroughly reviewed at ESA level during a mission requirements review.

Earth Explorer 9 call

The first issue of the call for the Earth Explorer Mission EE-9 in November 2015 did not result in the selection of mission concepts for preparatory activities at Phase-A level, primarily because none of the proposed mission concepts met the strict programmatic boundary conditions. ESA published a revised Call for Proposals for EE-9 on 13 December, with adapted boundary conditions. Proposals are due in June 2017 and two candidate mission concepts will then be selected to enter Phase-A in November 2017.

Meteorological missions

MetOp

The three MetOp polar-orbiting meteorological operational satellites, developed by ESA, constitute the space segment of the Eumetsat Polar System. MetOp-C has been removed from storage and the payload module and service module have passed separate thermal vacuum tests. The satellite is on schedule for launch in October 2018.

MetOp-SG

The second generation of MetOp satellites (MetOp-SG) is planned to continue and enhance the observations provided by the current, first generation of MetOp satellites. It will constitute the space segment of the second generation Eumetsat Polar System. MetOp-SG will consist of two series of satellites (A and B) and will provide the operational meteorological observations from polar orbit from the early 2020s to the mid-2040s.

The build-up of the industrial consortium in accordance with ESA's Code of Best Practices is virtually complete. The Phase-C/D price conversion process was completed and the updated contractual documents signed. The lower level subsystem and equipment design phase has been taking place, with more than 100 reviews held across the project.



The payload module of MetOp-C ready for testing at ESA's ESTEC technical centre in the Netherlands

MTG

The Meteosat Third Generation (MTG) satellites will ensure continuity of operational meteorological and climate data from geostationary orbit after completion of the ongoing Meteosat Second Generation missions. MTG-I (Imager) provides enhanced visible and near-infrared imagery and a new Lightning Imager. MTG-S (Sounder) provides a new infrared sounding capability from geostationary orbit and will also accommodate the Sentinel-4 UVN spectrometer.

The Critical Design Review (CDR) process is well advanced, with the majority of unit and equipment level reviews already completed. Preparations for the platform and instrument CDRs, to be held in 2017, are well advanced. Overall, the satellite (MTG-I and MTG-S) detailed designs are now complete with good predicted performance. With this progress have come some significant challenges, particularly at subcontractor level, where a combination of technical difficulties and capacity limitations has been encountered.

Other missions

SAOCOM-CS

SAOCOM-1 is an Argentinian (CONAE) polarimetric L-band SAR mission with two satellites for launch in the 2018–20 period. Following a free launch offer by Argentina, ESA and a wide community of scientists and firms defined a small companion satellite, the aim of which was to fly in formation with SAOCOM-1b to provide the first ever L-band bistatic SAR measurements from space for new applications.

Following a successful Phase-A/B1 that also de-risked the mission, Phase-B2 of SAOCOM-CS was carried out after the contract for the full procurement (B2/C/D/E1) had been negotiated. However, due to the under-subscription of Earth Observation Envelope Programme EOEP-5 at the Ministerial Council 2016, and the fact that specific contributions from interested Member States did not materialise, the SAOCOM-CS project was stopped after the completion of Phase-B.

Seosat/Ingenio

Seosat/Ingenio is a Spanish national mission developed by ESA on behalf of Spain's centre for industrial technology development, CDTI. It is planned to provide metric resolution multi-spectral land optical images. The primary payload flight model has been integrated and the pre-environmental tests are ongoing. The platform flight model is integrated and awaiting the primary payload delivery expected for mid-2018. The ground segment has performed the Critical Design Review and is being integrated.

Other activities

Ground Segment

The new Ground Segment Evolution Strategy was advanced by securing more than a €100 million at Ministerial Council 2016 for the implementation of Earth Observation Exploitation Platforms. These platforms and their interoperability will facilitate the access to Earth observation data dramatically. This approach also leads to the preparation of a Data and Information

Access System as part of the Copernicus Programme in close cooperation with the European Commission.

Climate Change Initiative (CCI)

The programme progressed according to plan during 2016. All projects have started Phase 2 of their Essential Climate Variable (ECV) production schedule. Two new activities have started that facilitate access to, and use of, ECV data products: the CCI Open Data Portal and the CCI Tool Box. In addition, 13 grants have been awarded to early career scientists for two-year projects that involve the novel use and exploitation of ECV products. CCI projects have been strongly encouraged to make their products available to Obs4Mips and support has been provided for this when needed. Nine CCI data sets have been approved by the World Climate Research Programme – Data Advisory Council. Eight of the ECVs within the CCI programme have been included in the Copernicus Climate Change Service call, giving strong opportunities for these projects to be incorporated in an operational context.

Earth Observation Data Exploitation

The ESA Living Planet Symposium was held in Prague, Czech Republic, from 9–13 May and opened by the ESA Director General. It was the largest annual Earth observation event organised in Europe with more than 3000 participants from 75 countries representing academia, industry and the public sector.

Science Element highlights include evidence of the largest inland expansion of supra-glacial lakes in the 21st century in northeast Greenland, a new climatology of ocean CO₂ fluxes exploiting satellite and in situ data, the first estimates of ocean acidification parameters by satellite, and the first estimates of mantle conductivity exploiting the geo-magnetic signal of tides as observed by Swarm. In addition a web story on Sentinel-1 results entitled 'Satellites confirm sinking of San Francisco tower' was the most viewed Earth Observation web story of the year and the first Sentinel-2 crop type mapping was released for the entire Czech Republic.

The User Element continued its engagement with user communities, promoting and demonstrating the benefit of Earth observation including participation at UNFCCC COP-22 in Marrakech. The Value-Adding Element, began a new phase of activities with the key international financing institutions in three thematic domains: urban development, agriculture and rural development, and water resources management. In addition, EOP signed a five-year Memorandum of Intent with the Asian Development Bank for strategic collaboration on Earth observation.

Campaigns

ESA airborne and *in situ* campaigns were an essential activity in 2016 for independently verifying the performance and accuracy of data and products from ESA satellite missions in orbit, with in situ, ship-based and aircraft-instrument measurement activities performed in support of SMOS, CryoSat and Copernicus Sentinel-3. Meanwhile, campaigns also played an essential role in preparing future Earth observation satellite missions such as the ADM-Aeolus Doppler wind lidar Explorer mission. A highlight of 2016 was the WindVal campaign flown in the region of Iceland through a partnership between ESA, DLR and NASA

to prepare for commissioning phase calibration and validation. These data are being used to test methods for validating the satellite data, and to help calibration/validation users prepare with simulated products.

CEOS/GEO

During its chairmanship of the Strategic Implementation Team of the Committee of Earth Observation Satellites in 2016, ESA aimed to foster the exploitation of Copernicus data, in particular from the Sentinel missions, to better address new emerging societal needs, especially in relation to climate, disaster risk reduction and sustainable development goals. These strategic priorities were also proposed to the Group on Earth Observations (GEO) and endorsed at the 2016 GEO Plenary.

International Charter 'Space and Major Disasters'

The International Charter 'Space and Major Disasters' continued to provide rapid access to Earth observation data in support of disaster relief worldwide. It was activated 36 times in 2016, with five activations relating to Hurricane Matthew.

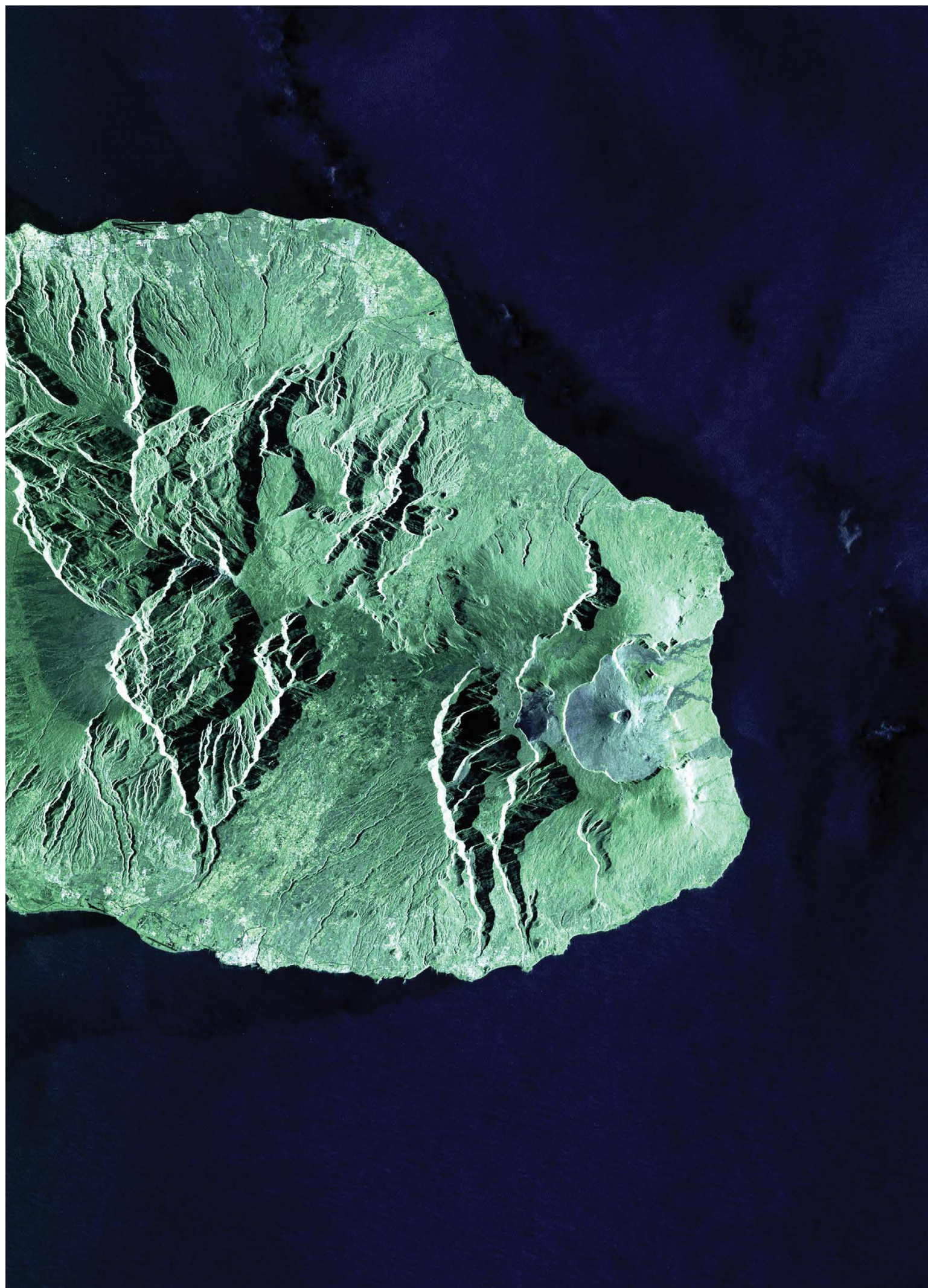
esrin

The ESRIN anti-seismic consolidation project has been ongoing with the completion of the childcare and gym facilities (Building 15). The tender process for the consolidation of B1 and B12 was also completed with works planned to commence in 2017. Owing to the seismic activity experienced in Italy in 2016, the seismic consolidation project timeline was reviewed and updated. This involves overlapping the consolidation of the B1 and B12 consolidation to speed up the completion of the overall project.

At the beginning of 2015, ESRIN's Energy Management System was certified according to ISO 50001:2011. Regarding environmental performance, in 2016 ESRIN reduced yearly overall energy use by 11%. The yearly energy usage reduction from the 2010 baseline is 32%.

ESRIN 2016 CO₂ emissions went down by 12%, with a yearly reduction from the 2010 baseline of 34%. Thanks to ESRIN environmental improvement initiatives, 1,480 tons of CO₂ have not been released into the atmosphere since 2010.

The ESRIN Host Agreement, signed in 2012, modified by notes verbales in 2015, was ratified by the Italian Parliament in August. Further to the exchange of the instruments of ratification and approval it entered into force on 6 October. The main purpose of the new Host Agreement, besides aligning the provisions with those of other international organisations in Italy, is to make ESRIN more attractive for ESA staff members and for ESA activities. By the end of 2016, ESRIN had 70 staff enjoying diplomatic privileges and immunities.





telecommunications & integrated applications

A year of successful delivery was reflected by the Member States at Ministerial Council 2016 in Lucerne, committing their largest ever subscriptions to the Directorate of Telecommunications and Integrated Applications' (TIA) programme of Advanced Research in Telecommunications Systems (ARTES).

The subscription of €1.2 billion recognises the effectiveness of ARTES as a mechanism for using innovation to strengthen the competitiveness of European and Canadian industry in the global markets for satellite communications and downstream applications. This secures economic returns of between 12:1 (direct returns) and 21:1 (including indirect returns) for every euro invested.

To this end, TIA operates from four sites (ECSAT, ESTEC, Headquarters and Toulouse), while the many sub-elements of ARTES are grouped into three main areas of activity: support for the Core Competitiveness of European space industry; Public–Private Partnerships; and Business Applications, plus the future preparations and ScyLight elements.

Core Competitiveness

ARTES Core Competitiveness was approved by the Joint Communications Board in September, providing a single framework that allows industry to start with an idea and develop it through to the introduction of new products and services to the market. There are two components: Advanced Technology; and Competitiveness and Growth. Around €90 million in funding was awarded during the year for new contracts, while the number of running contracts under Core Competitiveness exceeded 300 across 205 companies. This involved a total ESA price of €330 million, corresponding to more than €535 million of industrial costs.

The flexibility and streamlined process of Core Competitiveness enables ESA to support disruptive initiatives in areas such as 5G, megaconstellations and very high throughput satellite systems.

ScyLight

The Secure and Laser Communication Technology element complements the European Data Relay System (EDRS) and GlobeNET, enabling ESA to have optical communication technologies implemented as a standalone programme. Implementation is focused on key topics such as laser terminal evolution (for GlobeNET) and quantum key distribution.

Sentinel-1A radar satellite test image taken on 26 May 2016, recorded and relayed to Earth by EDRS-A on 31 May 2016 via laser

Public-Private Partnerships

ESA's Public-Private Partnership (PPP) mechanisms leverage public funding with private investment and ensure that risk in new ventures is allocated to those best able to handle it. By the end of the year, ESA had 15 PPPs in place or approved, including both ESA-initiated and industry-initiated examples, the latter conducted through the ARTES Partner programme. Between them, these PPPs involve all of the European satellite operators and primes.

Progress on the ESA-initiated PPPs is as follows:

EDRS

The first node of the European Data Relay System, or 'SpaceDataHighway', was launched aboard a Eutelsat satellite at the end of January. Commercial services started in November, linking EDRS-A with the Sentinel-1A satellite of the European Copernicus system. Mating of the platform and payload for the next satellite in the system (EDRS-C) was also completed for launch in 2017. Evolution into a full global system is planned through a follow-on programme, GlobeNet.



EDRS-A launch. The first laser node of the European Data Relay System lifted off from Baikonur, Kazakhstan, on a Proton rocket on 29 January

SmallGEO

The year that started with the launch of EDRS-A ended with the final testing and preparation for launch of the Hispasat 36W-1 satellite, the first use of the SmallGEO platform that was developed to address the market for satellites under three tonnes. The launch of the satellite by Soyuz from Europe's Spaceport in Kourou, French Guiana, is scheduled for early 2017.



A packaged SmallGEO/Hispasat 36W-1 is seen in the Antonov plane in Kourou, French Guiana

Neosat

The Next Generation Platform activities included important steps towards their objective of supporting the development and demonstration in orbit of new satellite platform product lines for the 3–6 tonne market. This included progress on intermediate design reviews for Spacebus NEO with Thales Alenia Space, and Eurostar NEO with Airbus Defence & Space. Orders were received during the year and the first launches for both types are scheduled to take place by 2019.

SAT-AIS

Microsatellites equipped with Automatic Identification System (AIS) receivers enable tracking of seafaring vessels beyond the reach of coastal radio stations. A new LuxSpace platform, Triton X, was supported at the Ministerial Council 2016. A first launch of Norway's NORSAT 1 carrying an AIS receiver is planned for 2017.

Large Platform Mission

In May, ESA reached agreement on a three-year extension of operation for the Test and Demonstration Payloads hosted on Inmarsat's Alphasat satellite, reflecting the flawless in-orbit performance of the spacecraft and its missions over the past three years.

Progress on industry-initiated ARTES Partner projects is illustrated in the list below, which is not exhaustive.

Electra

Under development for SES by OHB, this new satellite will save fuel by using full electric propulsion for orbit transfer and station-keeping. It will deliver communications capabilities similar to those of larger satellites, yet with a launch mass below three tonnes. Phase-B2/C/D/E1 was signed in February 2016 and the system design consolidation review was completed, with a launch planned for 2021.

INDIGO (Intelsat Newtec DialoG Open System)

The contract between ESA, Intelsat and Belgian company Newtec was the first aimed at delivering ground segment innovations, in this case for the Intelsat EpicNG constellation of high-throughput satellites.

Quantum

The most flexible payload yet designed, this in-orbit reprogrammable 'chameleon' satellite involves a PPP among ESA, Eutelsat and Airbus Defence & Space. The satellite Preliminary Design Review was completed in July and Eutelsat's baseline design review in December.

ESA's GOVSATCOM Precursor

This programme helps to prepare for the overall European Governmental Satellite Communications programme that is led by the European Commission and European Defence Agency. ESA's precursor activities were effectively launched during a conference at ESA Headquarters in April and project submissions the following month provided the basis for dedicated Member State support at the Ministerial Council 2016.

Iris

This project is developing the air traffic management (ATM) component of the Single European Skies ATM research programme (SESAR). In September, ESA signed a Memorandum of Cooperation



Florian Guillermet, Executive Director of the SESAR Joint Undertaking, and Magali Vaissiere, ESA's Director of Telecommunications and Integrated Applications, sign the Memorandum of Cooperation related to air traffic management

with the SESAR Joint Undertaking. The project involves a consortium of more than 30 organisations led by Inmarsat and the first flight trials were announced at the end of the year. ESA and the SESAR Joint Undertaking signed a memorandum of cooperation on 15 September 2016 to work more closely in the modernisation of air traffic management through satellite-based communications.

ICE

Expanding the range of mobile satellite services with Inmarsat at both L-band and Ka-band, ICE Phase 1 is developing real-time L-band services, while ICE Phase 2 was approved at Ministerial Council 2016 and aims to implement the complete system and service platform, using an open architecture with standardised interfaces to ensure an easy entry point for application developers.

Business Applications

In July, ESA issued a concept note on space-based applications, describing a new approach in which commercial applications using any space assets (Earth observation, navigation or satellite communications, singly or together) are addressed by the Integrated Applications Programme (IAP). For commercialisation and market uptake, such activities have the collective title Business Applications and are managed by TIA. Earlier stage activities (i.e. product driven research and pre-commercial development) are undertaken within the respective Directorates of TIA, Earth Observation and Navigation.

This approach provides greater flexibility in responding to market opportunities: 61 new activities were started during the year and just over two thirds of companies that signed contracts for IAP activities were SMEs. Outreach was boosted by the nine IAP



Monitoring bridges from space: the GeoSHM service monitors two giant bridges in China as well as the Forth Road Bridge in Scotland shown here (Forth Road Bridge)

Ambassador Platforms (APs), including an event with AP Norway in April and the launch of the new AP Italy in November. Sector workshops covered themes such as waste management and Alpine applications (with Eurisy), while the largest event was the 6th annual ARTES Applications Workshop, hosted in Brussels by BELSPO.

A new class of activities called Kick-start was defined to attract new players. Initiated by partners from industry, operators or public institutions, the activities will follow thematic calls for ideas or opportunities initiated by ESA, with the first calls scheduled for 2017.

In November, Seraphim Capital launched the first Venture Fund dedicated to space companies, the 'Seraphim Space and Special Situations Fund'. They are working closely with ESA to identify promising investment opportunities among both upstream and downstream companies across the space sector.

Communications and Promotion

Besides the telecom pages of the main ESA portal, the ARTES programme reaches out to industry and user communities through its dedicated websites for mainstream telecoms and applications respectively, artes.esa.int and artes-apps.esa.int.

During 2016, the sites were visited by 51 000 and 40 000 users respectively, achieving a combined total of more than 489 000 page views. They published a total of 100 news and events articles, 192 project web pages and 130 funding opportunities during the year.

Efficiency, ISO and Management Actions

In July, TIA's Quality Management System (QMS) passed a recertification audit, which was held under a new standard, ISO9001:2015. Three additional internal audits were held and 59 TIA staff were trained in QMS, reinforcing the adoption of ISO processes across the directorate. This is reflected in a risk management process that provides monthly reporting and decision-making at Director and Head of Department level.

ecsat & the harwell space cluster

The Director of TIA is also Head of Establishment at ECSAT in Harwell, where there is a fast-growing space cluster that welcomed its 65th space company at the end of the year. The majority of personnel at ECSAT are from TIA but it is also home to the ESA Climate Office and some activities of HRE and TEC. In June, ECSAT hosted a joint OECD-ESA event on innovation drivers in downstream space activities. The following month ESA Director General Jan Wörner inaugurated the ESA/RAL Advanced Manufacturing Lab, which is hosted by the UK's Science and Technology Facilities Council. The laboratory assesses new material processes, joining techniques and 3D printing technologies for applications in space.





galileo & navigation-related activities

In 2016, the Galileo Programme achieved a number of important technical and programme milestones. The European Commission declared the provision of Galileo Initial Services (15 December); the European GNSS Agency (GSA) and ESA signed the Working Arrangement covering execution of the Galileo Exploitation Phase for the 2017–21 period; and the EC and ESA signed an amendment to the Galileo Deployment Agreement covering the remaining satellite deployment activities, including the satellite Batch 3 procurement.

Galileo In-Orbit Validation Phase

In the context of the Galileo In-Orbit Validation (IOV) Phase, the Executive proceeded with the tasks required to achieve formal closure of all industrial contracts and initiated the transfer of assets to the GSA.

Galileo Full Operational Capability Phase

The in-orbit testing of spacecraft GSAT 0210/0211 launched in May has been completed. Four Galileo spacecraft – GSAT 0207/212/213/214 – were launched by Ariane 5 on 17 November (flight VA233). Respective in-orbit tests have started to allow a commissioning and entry into operation in 2017.

Concerning Full Operational Capability (FOC), the FOC₁ system qualification activities (based on 18 satellites, including four IOV satellites, and the associated ground segment control and mission capabilities) have been carried out, for handover to the GSA. The system performance during 2016 has been in line with the key performance indicators defined to support Initial Services. The Initial Services Validation campaign has been completed. The target definition and performance of Open Service has been achieved. The Ground Mission Segment has been upgraded since the beginning of August to also support the navigation message generation of satellites GSAT 0201/0202 in elliptical orbit. Inclusion of those two satellites in the operational constellation is planned for 2017.

Amendment No.2 to the Galileo Deployment Delegation Agreement between ESA and the EU has been finalised and adopted by Council at its 12–13 October meeting, and was signed by ESA and the EU on 26 October. Amendment No.2 was needed to establish the necessary financial and legal

Four new Galileo satellites were simultaneously launched into operational orbit on an Ariane 5 ES on 17 November

framework in a timely manner allowing for the procurement of Batch 3 of Galileo satellites to proceed.

EGNOS Exploitation Phase

The EGNOS activities executed in the frame of the EC/ESA Delegation Agreement were mainly related to the close out of the Framework Contract 1 (FWC1) and the execution of the new FWC2 work orders, devoted mainly to the resolution of obsolescence and the modernisation of the EGNOS infrastructure.

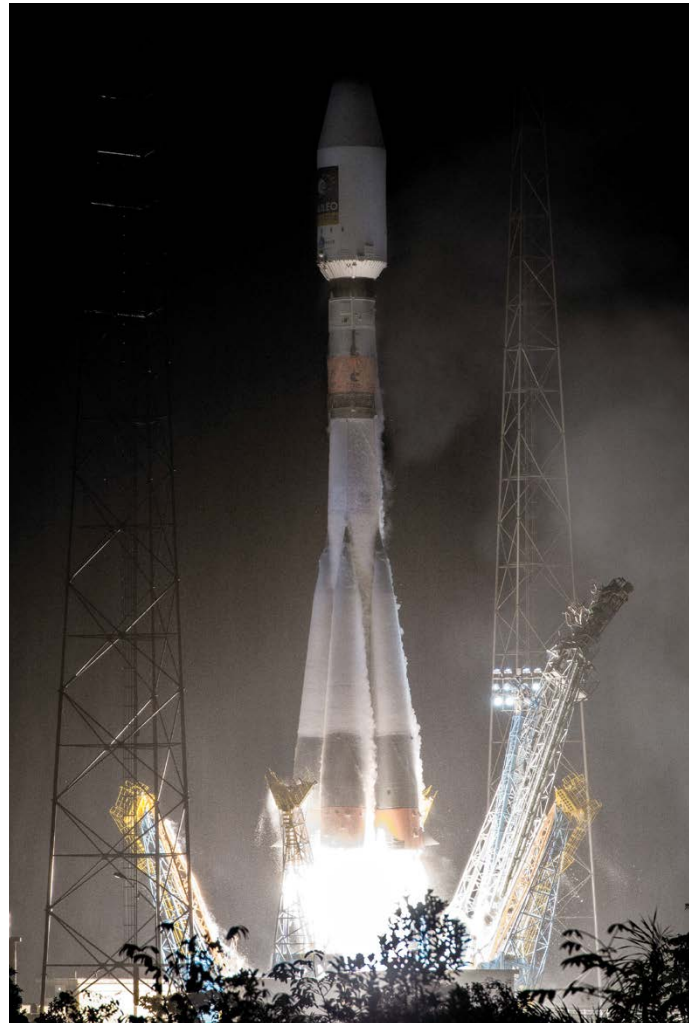
Activities undertaken for the GSA in the frame of the EGNOS Working Arrangement were mainly devoted to preparation for the procurement of the future system release V2.4.2 and the new generation EGNOS V3.

European GNSS Evolution Programme (EGEP)

The implementation of the EGEP research and technology development activities has proceeded according to plan, closing most of the EGNOS-related activities, and providing key enabling technology pre-developments in preparation for Galileo Second Generation (G2G).

The EGNOS-related activities carried out under EGEP concentrated on activities for EGNOS V3 Phase-B2 Definition and were completed in 2016.

Concerning G2G system and segment Phase-A developments, a major achievement was the preparation and start of the system Preliminary Requirement Review (PRR) activities closing the Phase-A developments, and targeting a PRR board meeting for early 2017. Noticeably, the scope of the review was extending beyond the scope of project activities, comprising programme-level considerations such as an EC-produced EGNSS evolution



Europe's 13th and 14th Galileo satellites (2010 and 0211) lifted off on 24 May from Europe's Spaceport on a Soyuz launcher (ESA/CNES/Arianespace)

High-Level Definition document (eHLD). The G2G PRR also involved representatives from EU member states/ESA Member States as board members or as reviewers in both system and mission/programmatic panels.



Europe's 13th and 14th Galileo satellites are loaded into their capsule fairing ahead of launch on 24 May (ESA/CNES/Arianespace)

EC/ESA H2020 Delegation Agreement

The Delegation Agreement between ESA and the EU on the implementation of Horizon 2020 Satellite Navigation Programme (HSNAV) proceeded according to plan with the implementation of the H2020 WP2015 activities and the approval of the H2020 WP2016 aimed at G2G Phase-B System and Segment developments.

Negotiations concluded with an agreement between ESA and the EU on ESA remuneration under the H2020 Delegation Agreement, with the objective of ensuring the required level of resources for the implementation of ESA technical and management activities in relation to HSNAV.

Navigation Innovation and Support Programme (NAVISP)

The NAVISP Programme was subscribed by ESA Member States at the Ministerial Council 2016. Initial steps have been

taken towards the preparation of the first NAVISP workplan concerning its element 1 related to innovation activities in satellite navigation.



ESA signing Galileo agreements with the EC and EU GNSS Agency on 15 December 2016. From left, ESA Director General Jan Wörner; Lowri Evans, EC Director-General of the Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs; Carlo des Dorides, Executive Director of the European GNSS Agency





space transportation

Launcher exploitation programmes

Ariane 5 exploitation

Seven Ariane 5 missions were performed in 2016, placing in orbit four Galileo European institutional satellites (EU) and 10 commercial satellites, with a record performance reached of 10 722 kg (separated payloads and carrying structures).

These achievements are the results of several activities put in place for the maintenance of the qualification status of the Ariane 5 Launcher System, which are performed in the framework of the Ariane ARTA and LEAP programmes.

The ARTA 6 hot-firing test of a solid rocket motor took place on 9 September at Guiana Space Centre; all the objectives of Level 0 exploitation phase were reached.

Concerning the Vulcain 2 ARTA 10 tests, 15 tests were performed in total, instead of the nine initially planned, due to various anomalies; the engine testing was completed and lessons learnt analyses shared. Several pyrotechnics equipment test campaigns were also performed.

A complementary Critical Design Review was completed for the new 'out of autoclave' fairing in July.

With a view to enabling increased Ariane 5 Geostationary Transfer Orbit launcher performance beyond 2018, additional complementary development activities have been initiated in the frame of the Performance Improvement and Competitiveness initiative and the preliminary design phase was concluded in October. The additional subscriptions necessary for covering industrial work were confirmed at the Council meeting at ministerial level in December.

The demonstration of compatibility between the Ariane 5 ES Galileo launcher and spacecraft was achieved in August by the 'Galileo FOC Space Segment Delta-Qualification to Ariane 5 Mechanical Environment' Review. The launch took place on 17 November.

An authorisation to proceed funded by the Science Programme for the adaptation of Ariane 5 to the James Webb Space Telescope launch was awarded to industry to allow a commencement of the activities in October.

Vega exploitation

Two Vega launches were performed in 2016, launching six commercial payloads.

Ariane 5 flight VA233 lifts off from Europe's Spaceport in Kourou, French Guiana, on 17 November



An ARTA-6 solid-propellant booster test-fired at Europe's Spaceport in Kourou, French Guiana, on 6 September

The full exploitation of Vega by Arianespace started with the launch of VV07 in September.

The last step of the Vega qualification process, the Post-VERTA Qualification Review step 2 took place in September and concluded that all objectives of the review were achieved. All the LEAP Vega classical and Ministerial Council 2016 planned activities have been committed.

A new rider was signed for the exploitation accompaniment activities with the Vega Launcher System prime contractor ELV, to be implemented through Work Orders.

Europe's Spaceport, the Guiana Space Centre

Eleven launch campaigns (seven Ariane 5, two Vega and two Soyuz) were performed at the Guiana Space Centre in 2016, to orbit 28 payloads (four of which were auxiliary payloads).

The initiatives aimed at improving the geo-return of Member States at the Guiana Space Centre have continued. In particular, following the positive reception of the 2015 event, Industry Days were held in Kourou and in Paris in October.

Launchers development programmes

Ariane 6

Consistent with the declaration on the programme for Ariane and Vega development, the states participating in the programme agreed to conduct a Programme Implementation Review (PIR) in 2016, on the basis of a report on the implementation of the Ariane 6 and P120C elements.

Regarding the implementation of Ariane 6 Launcher System development, from the technical (specification, design, industrialisation) and programmatic (processes, organisation,

schedule and cost) points of view, no showstopper was identified at Maturity Gate 5 (formerly Launcher System PDR), concluded in early June, and that was an input for the PIR.

Regarding the Ariane 6 Launch Base, acceptance of the earthworks activities took place in May. The Launch Range Adaptations PDR and Launch Base PDR were completed. Contracts for the ELA4 Infrastructures and Control Bench development were signed in July, works on the launch pad began at the end of July. The Industrial Critical Review took place for the structure relevant to the mobile gantry, the BAL and the water systems. An industrial PDR took place for mechanical systems (table, deflector, pallets and mast).

Concerning the Launch System Architect activities, the Launch System Preliminary Design Key Point took place in June. The Ariane 6 High Level Requirements and Mission Requirements Documents have been updated.

The PIR took place during the second half of the year and was concluded with a positive decision to proceed with Step 2 activities during the November Council. Consequently, the signature of contract instruments releasing full Step 2 activities for the Ariane 6 launcher and launch base took place on 9 November.

The Memorandum of Understanding between ESA, Arianespace SA, Airbus Safran Launchers SAS, ELV SpA represented by Avio SpA and CNES regarding the exploitation of Ariane 6 and Vega Launch Systems, was signed on 17 November. At the Ministerial Council 2016, some adjustments were made to the subscriptions in the Ariane 6 Element better to reflect the forecasting of geo-return. The level of the uncovered part remains unchanged

Vega C

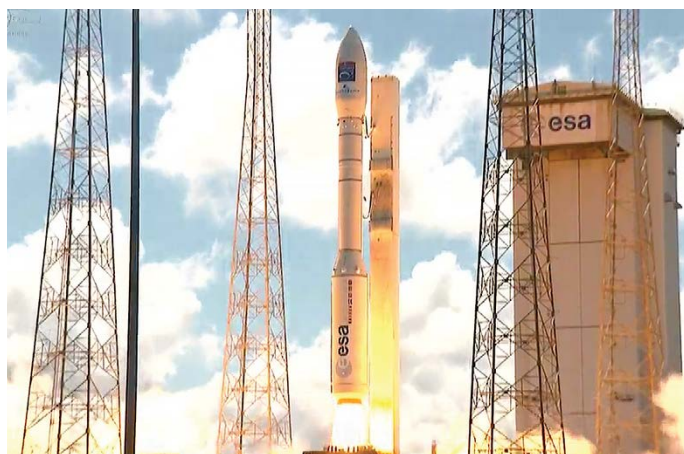
Following the performance increase change in the P120C SRM, additional activities were required regarding the Vega-C Launcher System and Launch Complex, with an impact of approximately six months in the planning of the activities.

Safety preparatory activities have been completed and the safety requirements and safety process have been consolidated.

The Launch Complex Ground Support Means contract kick-off was performed in June, with the agreement of minor modifications to the original proposal to comply with the new Vega-C+ configuration.

Vega Electrical Upper Stage (VEnUS) activities were initiated in October and are progressing according to plan. The activity aims to define an autonomous electric propulsion module enhancing Vega payload injection capability to non-Low Earth orbit orbits (medium Earth orbit, high Earth orbit, etc.), while exploring new launch services such as in-orbit satellite servicing and in-orbit debris removal.

Regarding the Small Spacecraft Mission Service (SSMS) initiative, the System Requirements Review was closed in December. The high-level programmatic objective of SSMS activities is to ensure an optimisation of the design for long-term Vega-C applications while enabling a first use on Vega in late 2018.



The second of 2016's Vega launches, VV08, carried a Turkish Earth observation satellite into orbit (ESA/CNES/Arianespace)

P120C

Following the decision of the Ariane 6 and Vega-C launcher system prime contractors on P120C performance increase, a P120C Solid Rocket Motor (SRM) Delta-Preliminary Design Review took place in June to address the evolved motor definition and the treatment of the actions and recommendations identified by the SRM PDR Steering Board. The Delta-PDR Steering Board authorised the entering of Phase-C of the SRM development. The Loaded Motor Case Delta-PDRs took place in July. The P120C Nozzle Delta-PDR was completed in September, with no particular technical issue raised. The P120C Inert Motor Case Delta-PDR was held on 7 December.

The contracts for P120C between the two launcher system prime contractors, ASL and ELV, and Europropulsion were signed in June.

Nammo was selected as supplier of the P120C Igniter, following a competitive tender process.

A key point on long lead items (LLI)/assets was held at the beginning of October, providing a good level of visibility on the status of LLI and assets of the P120C SRM. The P120C Thrust Vector Control/Thrust Vector Actuation System PDR was also conducted in September.

The BEAP adaptations PDR avionics part took place in July. Concerning the new booster case technology demonstration, the kick-off for Step 1.a activities took place on 1 June. The Pathfinder mandrel LLI key point was held in October.

Concerning the second production line, a Common Understanding between Germany, France, Italy and ESA on the way forward on Ariane 6 and P120C was signed on 3 November, and the Step1.b RFQ was released to industry in December.

Launcher Evolution Element

The Launchers Evolution Element (LEE) Initial Activities contract concluded with the final key point, held on 21 June, and the delivery of the LEE Programme File on 30 June. The LEE Programme File includes a work plan, estimated cost and planning for a modular demonstrator of reusability in the 2017–20 period.

Following LEE Initial Activities, preparations for follow-on activities in the LEE Bridging Phase started with a focus on system studies mostly oriented towards consolidating the requirements for the Prometheus engine. In the meantime, the Bridging Phase contract was signed and activities began with industry. The results from the LEE activities were considered when defining the activities proposed for the FLPP Period NEO Proposal.

Future Launchers Preparatory Programme (FLPP) Period 3

The programme incorporates the continuation of the current LEE and FLPP activities into a further New Economic Opportunities (NEO) period of FLPP.

During the Ministerial Council in December, different components and corresponding financial envelopes were approved and subscribed for FLPP Period NEO: FLPP NEO core component, Prometheus demonstrator component, LOX-methane test bench component.

FLPP Period-3 is proceeding as planned in the areas of integrated demonstrators and technologies.

The material preparation phase of Expander Technology Integrated Demonstrator (ETID-1) hotfiring test campaign has started at the P3.2 test bench, after the Critical Design Review was held on 27 November. The ETID-2 propulsion system concept phase was kicked off on 14 December. The turbine bypass valve and hydrogen chamber valve passed their respective MRRs on 24 and 25 October. On the ultra-low cost engine demonstration (Prometheus), the Concept Review of the LOX/Hydrocarbon combustion chamber was performed on 13 December. The contract was signed for 'Integrated Stiffened Cylinder' activity (aiming at low-cost upper-stage cryotanks by metal flow forming), performed in cooperation with NASA and Lockheed Martin. The FORC Fibre Optimised Reinforced Casing pressure vessel was manufactured and underwent non-destructive inspection. A Test Readiness Review was held in December.

Space Rider

The Programme for Reusable In-orbit Demonstrator for Europe (PRIDE) is now known as Space Rider. The high-level mission requirements target a reusable integrated space transportation system to be launched with the Vega-C launch system, with a payload mass capability higher than 300 kg for a multitude of orbits, altitudes and inclinations (including the ISS orbit), able to operate in orbit, deorbit, reenter, land on ground and be relaunched after limited refurbishment.

The industrial activities began in March. Phase-A activities concluded with the Preliminary Requirement Review collocation meeting held on 22–23 November. Phase-B1 activities were redirected to take into account of the integration of VegaC/AVUM as Orbital Service Module. Meanwhile, preparation of the decision on the programme's Step-2.1 (2017–19) continued and Step 2.1 was subscribed at Ministerial Council.





operations

Highlights

This was another record year for the Directorate of Operations, with four LEOPs (launch and early orbit phases) conducted from ESOC in just over three months: Sentinel-3A in February, ExoMars Trace Gas Orbiter (TGO) in March, Sentinel-1B in April and Galileo L7 in May.

In addition, two critical interplanetary operations were conducted in the autumn: the Rosetta end of mission with soft touchdown on the comet's surface and the arrival of the TGO at Mars with the delivery of the Schiaparelli landing demonstrator module. ESOC staff, in conjunction with France's space agency CNES, also carried out (from Toulouse) the first quadruple LEOP for the Galileo satellites launched on Ariane 5 in November. At the end of the year, OPS was controlling 20 satellites in flight, including three operated from Redu.

Space Situational Awareness

Provision of space weather pre-operational services to users continued throughout 2016. In June, the SSA programme organised a workshop at ESOC with European National Civil Protection Authorities (NCPAs) focusing on space weather risks.

The SSA Cost Benefit Analysis study was completed in September. A benefit-to-cost ratio of 6.25 was estimated for the SSA Space Weather services over the 2016–32 period.

The Phase-0 studies for the space weather mission to L1 and L5 points, were completed in November.

The Next Generation Radiation Monitor (NGRM) Proto-Flight Model (PFM) was ready for integration with the EDRS-C satellite. Implementation of the Service Oriented Spacecraft Magnetometer (SOSMAG), continued. It will fly as a hosted payload on board the Korean GEO-KOMPSAT-2A meteorological mission. The related ESA-Korean agreement was signed in March.

The focus of the Near-Earth Objects segment is the development of the first NEO Survey Telescope that is being built by a consortium led by CGS (IT). Development is progressing as planned. Another telescope, the first test-bed robotic telescope acquired first light at Cebreros and will serve as a test platform for the fly-eye telescope.

In the area of Space Surveillance and Tracking (SST), the first phase for the establishment of Expert Centres for passive optical and laser observations is nearing completion. Integration of the SST core software has been initialised, including functions for sensor scheduling, orbit determination,

Deputy spacecraft operations manager
Silvia Sangiorgi is seen working in the
Main Control Room on ExoMars 2016
launch day (ESA/J. Mai)



The test-bed telescope of ESA's Space Surveillance and Tracking segment of the Space Situational Awareness Programme at Cebreros, Spain

catalogue database and user services required by the SST community in Europe. A study focusing on the capabilities of the existing and planned radars of the European Incoherent Scatter (EISCAT) Association has been completed. A major goal of this study is the characterisation of the envisaged EISCAT 3D multistatic UHF patch antenna field for surveillance tasks in low Earth orbit.



Some of the Sentinel-3A project team working at ESOC during launch on 16 February



The new space debris facility located at ESA's ESOC mission control centre

Mission Operations

ESOC continued to perform high quality mission operations, controlling up to 20 satellites in flight, ranging from near Earth orbits to interplanetary ones.

High efficiency and flexibility were demonstrated by the operations centre by performing four critical LEOP activities with a separation between them of 3–4 weeks. The high workload was sustained without any impact on the usual high quality of the work. At the same time, it was possible to achieve cost efficiency especially on missions, like the Copernicus Sentinels, that involve a series of identical or similar satellites being operated.

The unique expertise that ESOC has gained in the past 15 years in the area of interplanetary mission operations allowed the centre to achieve another major success in the operations of Rosetta, when the spacecraft was directed to perform a soft touchdown on the surface of Comet 67P, with the scientific payload operated until the last few metres, and achieving a landing accuracy of 35 m at a distance of more than 700 million km.

The same expertise was used in the critical operations of Mars orbit injection for ExoMars TGO and the delivery of the Schiaparelli test lander. All activities were conducted successfully for TGO, which achieved the nominal orbit around Mars. For Schiaparelli, all operations were also successful, achieving perfect targeting and managing to retrieve and transmit to Earth all data produced in real time by the lander. Unfortunately, the lander's autonomous activities were not successfully completed and the module crashed onto the surface. However, the recovery of all data has allowed industry to identify the cause of the problem, making the test a success notwithstanding the landing failure.

Ground Segment Engineering

Through the improvement of modelling the Galileo satellites' orbit dynamics and the processing of Galileo multi-frequency observations, it was possible to improve the accuracy of the Precise Orbit Determination for the Galileo satellites by a factor of two. This resulted in an accuracy for the Galileo Geodetic Reference Frame generation that is the highest achieved by any of the Global Navigation Satellite Systems.

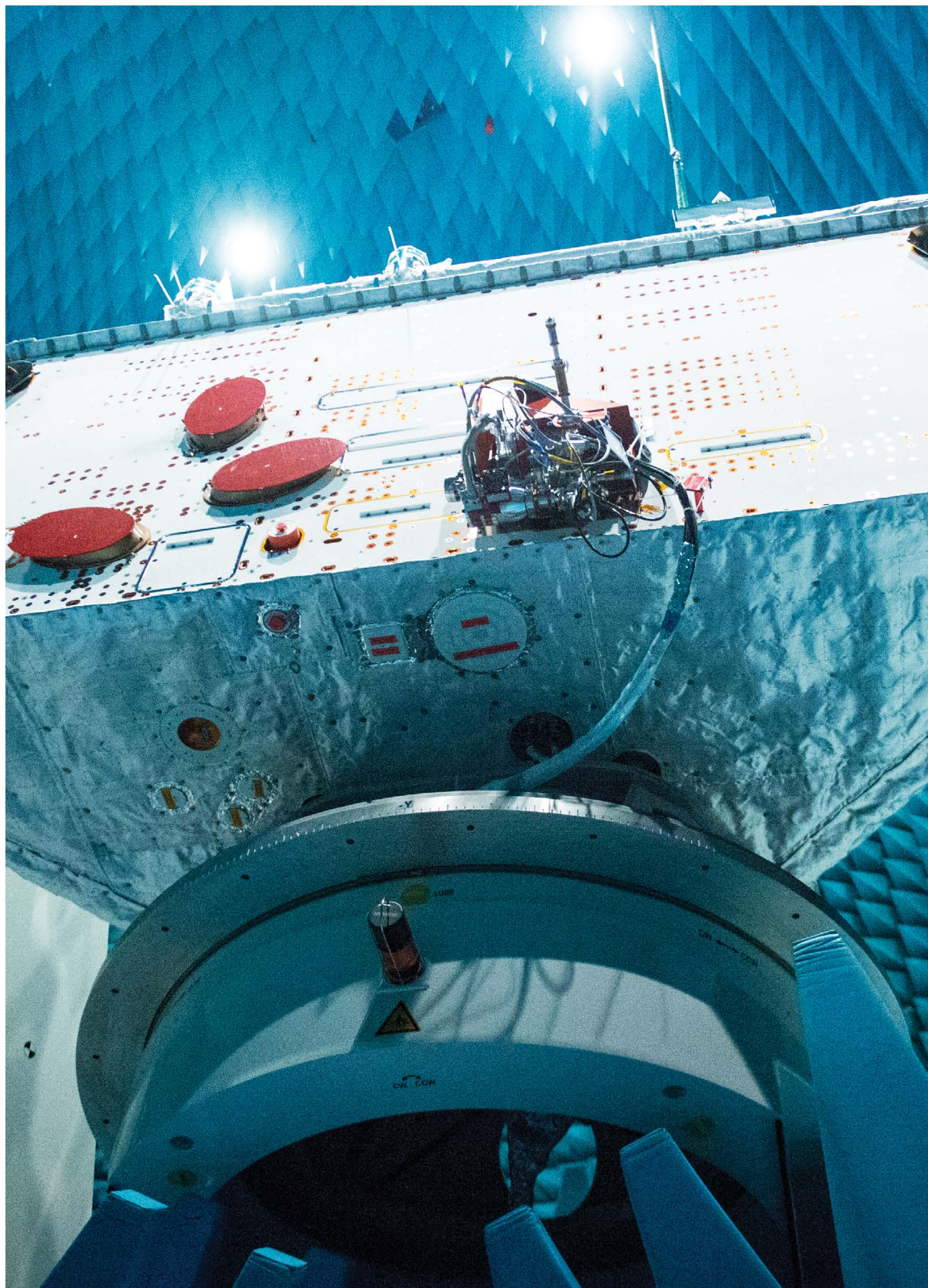
In the area of ground stations engineering, the Tracking, Telemetry and Command Processor (TTCP), ESA second-generation deep-space modem development, was concluded, and is ready for deployment to ESTRACK deep space stations.

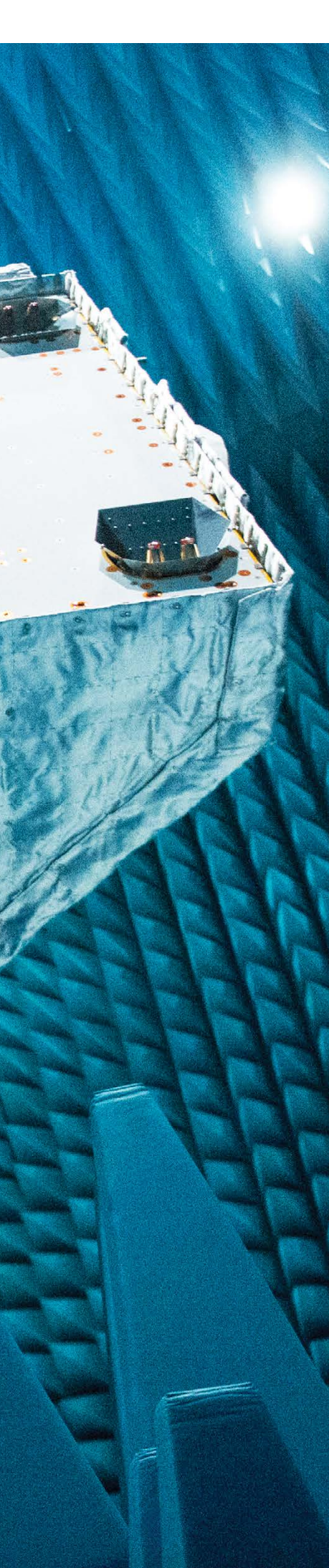
A new space debris operations room has been inaugurated, from which the 24/7 collision avoidance service is provided. The room is also the home for research and development activities in the domain, and serves as a hub for visitors and

decision-makers to familiarise themselves with the activities carried out by OPS in this area.

The European Ground Systems – Common Core, the European joint initiative to develop and exploit a common infrastructure for space systems testing and operations has progressed in the framework of the Phase-C/D contract. The preparation for the integration of such a system into the OPS infrastructure has started. The end-to-end ground segment reference facility, the environment for integration and testing of ground systems in a mission-operation representative environment, has been exercised in the framework of pilot projects.

OPS collaboration with TEC on the generation of the ESA UTC reference, also known as UTC (ESA/ESTC), resulted in a solution that is both accurate (nanosecond level) and robust. A request for registration of has been filed with the Bureau International des Poids et Mesures.





technical & quality management

ESA's future missions are made possible by its Directorate of Technology, Engineering and Quality, which oversees a portfolio of research programmes whose aim is to enable a seamless chain of innovation, producing a stream of enabling technologies and ground-breaking products.

General Studies Programme

The long-running General Studies Programme (GSP) continued to support the investigation and studies of new technologies and concepts, as well as all of the new ESA mission proposals up to Phase-A, strategic studies and long-term planning. The 'High Definition-GNSS' project analysed innovation in navigation, including extension of service to the poles, while there were also studies on asteroid exploration and exploitation, additive manufacturing and human lunar return. The GSP's Advanced Concepts Team has looked ahead further still, with research including a deep learning neural network for complex planetary landing and ray-tracing software to study possible signatures of the Milky Way's central black hole.

Technology Research Programme

This is the single programme engaging the largest number of SMEs across all Member States, adding up to one quarter of its annual budget. The long-running Technology Research Programme (TRP) remains ESA's innovation factory – serving all programme domains at lower technology readiness levels (TRLs). The 2017 TRP Work Plan was approved in November, based on IPC-provided information later complemented by Exploration domain activities, increasing its total to more than 220 activities in all.

General Support Technology Programme

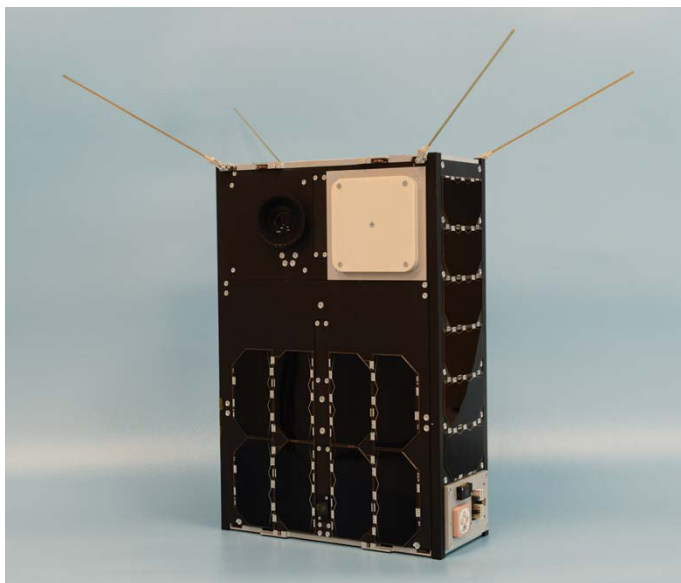
The Ministerial Council 2016 showed continuing confidence in the higher-TRL focused General Support Technology Programme (GSTP), with almost €450 million in new subscriptions, including support from the recently joined Estonia and Slovenia.

The Ministerial Council also introduced a new legal framework and updated programme structure for GSTP comprising three elements. Element 1 'Develop' covers future mission technology, ground applications and tools, its workplan including more than 260 activities overall (with more than 50 addressing the needs of the Cross-Cutting Initiatives); Element 2 'Make' develops technologies and products for commercial competitiveness and sustainability, co-funded with industry, research centres and universities; Element 3 'Fly' focuses on early in-orbit

ESA's BepiColombo Mercury Planetary Orbiter under test inside ESTEC's Maxwell chamber in Noordwijk, the Netherlands

demonstration of new technologies, future mission preparation and small missions.

Element 3, including the development of the Ops-Sat CubeSat, which passed its critical design review, and the initiation of two new missions: the demonstration of a mini Hall Effect Thruster with miniHETsat and new CubeSat mission GomX-4, in addition to existing CubeSats Qarman, PICASSO and Simba, plus mission design of the Hungarian-led RADCUBE. Proba-3's progress continues, on course for system Critical Design Review in 2017.



The ESA GomX-4B CubeSat

Technology highlights

TRP and GSTP results were proactively promoted through the first Coordinated Final Presentation Days and the new 'Shaping the Future' website.

Highlights included the hyperspectral Hyperscout payload for nanosatellites, operating in VNIR with a continuous tuning capability, and the ESA Micro Newton Thrust Balance for micronewton cold-gas thruster testing, developed in close association with the UK's National Physical Laboratory. A 5 m diameter Large Diameter Antenna was developed through TRP by a consortium led by HPS GmbH (DE) with Germany's Large Space Structures responsible for its finely knitted metal mesh reflector surface and antenna support structure. This achievement helps to fill a strategic gap in European capabilities.

Cross-cutting initiatives

These initiatives focused on particular themes of future interest, bridging different programmes, technical disciplines and technology levels. Advanced Manufacturing seeks to spin-in disruptive manufacturing processes and materials to open up industrial possibilities in terms of design freedom, optimised lead times and reduced costs, changing the way industry works and regaining lost European capabilities. In parallel and support to this initiative, the ESA/RAL Advanced Manufacturing Laboratory in Harwell has become operational.

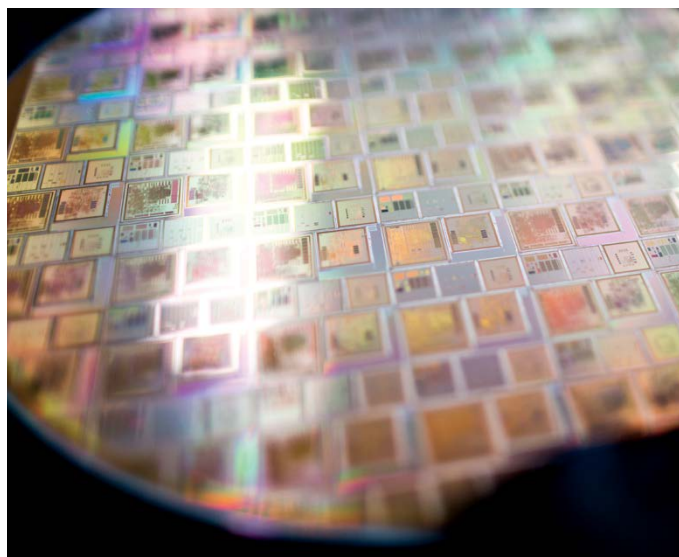


A very finely knitted metallic mesh on the top parabolic net of the 'double pantograph' large reflector demonstrated at ESTEC in November

The Clean Space initiative to decrease the environmental footprint of the space industry both on Earth and in space has continued its implementation, with the results of previous studies being translated to the GSTP work plan. As an example of progress, the modular Passive De-orbiting Subsystem to speed natural atmospheric decay after a satellite's end of life has reached TRL 5–6 with testing of an engineering model.

European Components Initiative

ECI held a two-day event at the Italian space agency ASI headquarters with more than 80 component suppliers and end-users to showcase the technologies and components developed since its 2004 inception. ECI is nearing the completion of its fourth phase and continues to add new parts to the European Space Components Coordination lists; in 2016 these included European MMICs for voltage control and wideband amplifiers, low and high power coaxial isolators for RF applications and a range of power MOSFETS in SMD packaging.



Multiple integrated circuits at the heart of Europe's space missions, etched together onto a single piece of silicon (ESA/G. Schoonewille)

Concurrent Design Facility

The Concurrent Design Facility has supported mission and system studies for a variety of ESA programmes while also preparing for the new Cross-Cutting Initiative, Design 2 Produce.

ESTEC Test Centre

A large number of satellite tests were completed within ESTEC's own testing centre. An additional seven Galileo satellites underwent acceptance testing, while the three BepiColombo modules continued their acceptance testing at system level, along with high-intensity Sun illumination tests for their solar array cells. Sentinel-2B achieved full acceptance.

Available resources have been deployed carefully to keep facilities operational and implement improvements where possible. Maintaining facility reliability is a priority, achieved through in-house engineering as well as industry support.

Technology coordination and planning

Regular meetings with Member State delegations (IPC-THAG), industry and other stakeholders took place through the European

Technology Harmonisation process. Updated European technical dossiers and roadmaps for eight topics: Avionic Embedded Systems, On-Board Payload Processing, Data Systems and On-board Computers, Microelectronics, Reflector Antennas, RF Metamaterials and Metasurfaces, Technologies for Passive millimetre and sub-millimetre Wave Instruments and Electric Propulsion Pointing Mechanisms.

2016's European Space Technology Master Plan, which included an introduction from both ESA's Director General and the European Commission Director-General for Internal Market, Industry, Entrepreneurship and SMEs, Ms Lowri Evans, was distributed to stakeholders. The Technology Advisory Working Group continued to hold regular meetings in conjunction with the IPC.

The TECNET end-to-end technology management process was revised to further improve its efficiency and effectiveness in coordinating ESA R&D. Meetings of the TECNET Chairs Forum supported the Adjudication Committee by analysing technology aspects of papers before they were processed.

Discussions continued with the European Commission on technology coordination with EU programmes. The EC/ESA/EDA Joint Task Force on Critical Space Technologies for European Non-Dependence worked with stakeholders to prepare a 2018–20 List of Actions.





resources management, corporate activities & external relations

Human Resources, Facility Management & Informatics

In 2016, special attention was given to leadership performance and development. A new 360-degree feedback tool and process has been designed and the first pilot was initiated, the aim being, over time, to address the entire managerial group at ESA.

A staff engagement survey was implemented, as a joint project with the Central Staff Association Committee, with a very high level of participation (82% of staff took part) and a high overall engagement score of 78%. This survey produced many interesting lessons and insights, notably on dialogue in the organisation and on improvements still to be made. The end of 2016 was devoted to analysing the results and drawing up an action plan.

During the year, ESA also focused attention increasingly on its external outreach as an employer. Changes have been made to our attendance at job fairs, while presentations and road shows have been organised in a number of our Member States, and our presence on social networks has been extended. ESA has also started a project to develop its employer branding and the first part of the project, consisting of surveys among the target audience, has been completed.

In 2016, ESA also took many opportunities to increase best practice exchanges with other organisations, both public and industrial. Our aim is to keep a constant eye on the state of the art and fine examples of human resources practice.

The year also brought significant changes to the working conditions of new staff members. Five years after the reform of the expatriation allowance, it was decided that staff members recruited as of January 2017 shall be entitled to family allowances under conditions that are in many family scenarios less favourable than in the past, but which will achieve savings to the Agency.

HR operations

With the objective of replacing outdated IT applications and increasing efficiency by rationalising and harmonising procedures, while providing full visibility and transparency towards staff and managers, the deployment of the first two modules of the new system hr-space (staff administration and recruitment) were prepared for deployment in early 2017.

Working arrangements

ESA has developed new working arrangements, in particular a new policy on teleworking, to provide more flexibility for staff to perform their normal duties outside Agency premises. The teleworking policy is one example of how ESA has responded to the wishes expressed by staff in the staff engagement survey concerning the need to balance work and private life. New rules were also introduced allowing staff members to take extended periods of paternity and parental leave in the case of multiple births.

Learning and Competency Development

Under the HR Competence and Policy Centre, different types of activities were conducted in 2016, from the implementation of a large range of training courses and programmes to the introduction of new measures aimed at further supporting competency development across the Agency.

In 2016, special efforts were made to redesign and implement a Newcomers Integration programme, to introduce a new System Engineering programme, and to review in depth the ESA leadership development programmes, around specific themes such as developing members of a team and leading innovation and change.

Pilot schemes have been started to evaluate how best to use modern technology to support learning, in particular towards the future generation of ESA staff, as well as to encourage knowledge transfer from senior to junior generations.

Young Graduate Trainees & Internal Research Fellows Programmes

The ESA Young Graduate Trainee (YGT) programme provides hands-on experience to young graduates from ESA Member States and Cooperating States. In 2016, out of the 3579 applicants to the YGT Programme, 90 candidates from 24 different nationalities received an offer following their interviews. In total, 179 YGTs were in post at ESA during the year.

The Internal Research Fellows Programme offers scientists and engineers the opportunity to carry out advanced research in a variety of disciplines related to space science, space applications and space technology. In 2016, the Internal Research Fellow Programme welcomed 20 new recruits from nine different nationalities. In total, 66 Postdoctoral Research Fellows were in post at ESA during the year.

National Trainee programmes

The National Trainee programmes for Germany, Luxembourg, Belgium, Portugal and Switzerland continued their activities during 2016. New agreements were signed with Estonia and Ireland, and the first national trainees from those two countries are expected to arrive in 2017.

Statistics

Staff

At the end of 2016, the Agency had 2285 staff in post, compared with 2266 at the end of 2015. There were 93 external recruitments.

Pensions

At the end of 2016, there were 1443 pension recipients, compared with 1398 at the end of 2015.

Member States	Hors Classe	A	L	B	C	Total Staff
Austria	1	37	1	2		41
Belgium	1	97		5		103
Czech Republic		5				5
Denmark		13				13
Finland		17		5		22
France	2	416	9	70	4	501
Germany	2	388	5	43		438
Greece		18		2		20
Ireland		21		6		27
Italy	1	396		23		420
Luxembourg		3				3
The Netherlands	1	63		12		76
Norway		23		2		25
Poland		14				14
Portugal		26		2		28
Romania		5		1		6
Spain	1	200		4		205
Sweden		42		2		44
Switzerland	1	27		1		29
United Kingdom	1	220	3	16		240
Total	11	2031	18	196	4	2260
Non-Member States						
Canada		24		1		25
Total		24		1		25
Grand Total	11	2055	18	197	4	2285

ESA staffing as at 31 December 2016

Estates & Facilities Management

Efforts to ensure that ESRIN is fully compliant with Italy's demanding anti-seismic buildings legislation continued at an even faster pace. Using any available original documentation from the construction of the buildings concerned, a mathematical analysis is being carried out. This involves a dynamic analysis of the structure following the specifications of applicable norms.

Owing to the age of the buildings on site, not all the structural drawings were available. Where the historical drawings are missing, a detailed investigation into the structure has been undertaken to replicate them. This requires extensive invasive and non-invasive investigations

to determine the structural elements of the buildings. Once all the information is collected, a mathematical simulation of the structure can be carried out to determine the seismic vulnerability of the structure.

To determine the characteristics of the materials used in the construction of the buildings, samples were collected, visual inspections undertaken and concrete compression and steel tensile tests performed.

Once the seismic vulnerability of a structure is identified, a number of different reinforcement solutions are considered to reinforce the structure to the highest possible requirements while minimising the impact on ESRIN day-to-day activities.

There is still much work to be done at ESRIN to complete the seismic reinforcement programme but it remains on schedule and cost. The ESRIN Estates and Facilities Management team have gained a vast amount of knowledge in the process, all of which augurs well for the future of the establishment.

Information Technology

In 2016, the IT Department released several new products and services for ESA and further developed new digital solutions. These new agile products and services directly address user needs in terms of effectively implementing the digital agenda.

hr-space

hr-space was developed to move critical human resources management functions and records from legacy paper systems to online. This system allows ESA staff to access their records, update their information and perform leave and other requests automatically. The project involved significant collaboration with HR to develop the product and related training.

esa-star

esa-star represents an evolution in the tendering process, by moving its legacy paper application system online and paving the way for an integrated, modern, public-facing portal for ESA's industrial partners. The IT Department is working to promote simplicity and user friendliness for online transactions with ESA.

easy-PSS

To complement the release of esa-star, a new system for online tender form management called easy-PSS has also been released to ESA's industry partners (PSS is an historical acronym and refers to ESA Procedures Specifications and Standards, to which is added an 'A' series of tables, or costing forms). The aim of this tool is to make the cost breakdown reporting for the tendering process more transparent and effective.

ESA's document management

Several steps have been taken to improve ESA's document management. SharePoint products (esateamsite) have been made available to all teams and offer tailored solutions for the Agency's varied needs. The IT Department has proactively increased awareness of the best document-handling procedures to improve ESA efficiency.

ESA eID

Online document signing has been simplified and improved by the introduction of ESA eID. This digital identification certificate is a legally binding, paperless signature. This will allow ESA to improve document workflows and international paperless collaboration.

esabox

The esabox file-sharing tool was released for all ESA users. It provides flexible, secure and compliant solutions for global collaboration and interaction. The tool is available to share with internal and external partners and is designed to allow easy file sharing with ESA's external partners.

digital security

The efforts of the IT Department have also included major digital security actions and ensuring ESA is a key actor in inter-Agency security collaboration. ESA has remained digitally secure despite several global threats. The IT Department is providing secure, agile, and paperless solutions for an ESA committed to a digital future.

Procurement, Financial Operations and Legal Affairs

Industrial Policy

Earth Observation

The year has been mainly dedicated to the consolidation of the Consortium for MetOp, Biomass and Sentinel-5 and Sentinel-6 under the Best Practices scheme with several hundreds of competitive procurements.

Leonardo SpA (IT) was selected as Instrument Provider for the FLEX Mission (competitive ITT).

SAOCOM procurement was initiated and completed, however the decision was taken at Ministerial Council 2016 not to place a contract due to budgetary constraints.

Telecommunications and Integrated Applications

A very high pace of commitments was maintained for ARTES technology programmes and Integrated Applications. The negotiations for the Electra Development phase (PPP with SES Astra and OHB) were finally concluded and signed.

Navigation

Amendment#3 of the ESA-EC Delegation Agreement as well as the Working Arrangement (Frame Contract) with the European GNSS Agency (GSA) for the Galileo Exploitation Phase were negotiated and signed at the end of 2016.

Following a long procurement process, the Industrial contract for System and Service Support was signed with Thales Alenia Space (IT).

The Competitive Dialogue Procurement action for the Ground Segment of Galileo (Ground Mission and Ground Control) has been initiated with the pre-selection of tenderers and competitive dialogue phase. This will be continued for most of 2017.

The procurement of a third batch of eight satellites (and options for six more) was brought to completion with ESA's recommendation being made to the European Commission. The contract is expected to be signed in mid-2017.

Under the Delegation Agreement with the European Commission for navigation activities under H2020, many framework contracts were initiated, with a first set signed at the end of 2016.

The In-Orbit Validation (IOV) contract closeout process has been continued and preparation has begun for the handover of operations to the GSA (planned for mid-2017).

Launcher Programmes

ESA signed the riders to the contracts for the launcher system and the launch base on 9 November binding upon industry and the French space agency CNES for the full development of Ariane 6.

The Ariane 6 launcher system development with Airbus Safran Launchers (ASL), is valued at €2.4 billion, including the maiden flight of Ariane 6 in 2020. A specific characteristic of the contract is the agreement of the prime contractor ASL and its main subcontractors to invest up to €400 million in the project, thereby confirming the commitment of the industrial sector to this new launcher development activity. As a new feature, the design authority for the launcher is fully in the hands of industry, which is counter balanced by an increased warrant regime in favour of the Agency.

The second contract is between ESA and CNES, for the development of the Ariane 6 launch pad and integration facilities at Europe's Spaceport in Kourou, French Guiana. The contract value is €600 million and covers the costs of the new Ariane 6 launch base. The formal handover of the facility by CNES to ESA is scheduled for late 2019.

In addition, the Contract Change Notice to Rider 1 to the Vega Consolidation and Evolution Programme (VECEP) frame contract and Contract Change Notice 9 to Work Order 4 were signed on 6 December contracting fully the step 2 activities of both the P120C Solid Rocket Motor and the P120C Thrust Vector Control.

Science Programme

Following approval of the procurement proposal by the IPC in October, the Rider for the evolution of the Solar Orbiter Implementation Phase have been finalised with Airbus Defence and Space Ltd (UK) with the Rider due to be signed in early 2017.

Following approval of the procurement proposal by the IPC in May, the negotiation for the Solar wind Magnetosphere Ionosphere Link Explorer (SMILE) payload module was completed with two parallel contracts with Airbus Defence and Space SAU (ES) and QinetiQ (BE) signed in early 2017. Best Practices selection was carried out for the Euclid and JUICE Projects.

Human Spaceflight & Robotic Exploration

ExoMars 2020: The rider covering the complete Phase-C/D was signed on 16 December following the decision taken at the Ministerial Council 2016 in Lucerne and corresponding IPC approval.

IBDM Development for the Dream Chaser Space Transportation

System: The new programmatic scenario for IBDM was presented to and approved by PB-HME, and the corresponding Contract Proposal approved by the IPC on 23/24 November. The next steps are the finalisation of a revised agreement with the Sierra Nevada Corporation and the consequent finalisation of the contract with European industry in 2017.

Lunar Exploration: The PILOT and PROSPECT Phase-B+ Contracts were signed with, respectively, Airbus Defence and Space GmbH (DE) in June, and Finmeccanica SpA (IT) in April.

MPCV-ESM: The Contract Proposal for ESM-2 was approved by the IPC on 23/24 November, on condition that the commitment and signature of the contract Rider for FM2 would be subject to the subscription of the required funding to the European Exploration Envelope Programme by the Participating States at the Ministerial Council 2016. Following positive decisions at that Council meeting at ministerial level, the corresponding Rider was due to be signed in early 2017.

Finance, Planning and Control Operations

Financial Operations

Intense efforts were devoted in 2016 to certifying the financial statements and focusing on correctly managing and accounting the Agency's payments, some 75 000 transactions related to payments and mission settlements.

In the context of the Procure-To-Pay (P2P) project, a Work Instruction that sets out the implementing rules for the administrative instruction 'Corporate Policy on the drafting, implementation and execution of financial and payments related clauses in ESA contracts' has been issued. The purpose of the P2P project is to optimise and standardise related processes to ease timely payments to suppliers while ensuring full compliance with accounting standards, and further improve the efficiency of payment operations.

Corporate Planning, Budgeting and Controlling

Corporate planning and control played a key role in 2016 in proposing budgets that were commensurate with cost plans approved by the Programme Boards, in discussion with contributors and taking into account their funding capacity; and ensuring a balance between cash outflows and available resources while limiting the need to activate credit lines on outstanding contributions to just before call-ups.

The multi-year planning process allowed the Executive to plan and monitor resources and progress of programmes, projects and activities, and to take action in case of significant variances.

Thanks to the additional contributions provided by some Member States, and the continuous effort to optimise planning and monitoring cash disbursements, the Agency's financial outlook was considerably improved with respect to initial forecasts. The Division ensured a positive treasury

balance throughout the year while meeting all financial commitments with industry and other partners.

In addition, the Division made a significant contribution to the preparation of the Ministerial Council 2016 – internally, with the preparation and consolidation of plans, support to the definition of new programmes and activities and strategic financial advice to the Director General and Executive Board and externally, providing substantial support and advice to Member States and Participating States in relation to possible subscriptions, simulations and affordability requirements.

Progress was achieved in the development of an Agency-wide tool for planning and control with the conclusion of its 'Plan and Analyse' phase. Deployment of the tool is expected to occur in 2018.

Cost

ESA's total cost in 2016 amounted to €5497 million, of which €3976 million related to ESA programmes and activities and €1521 million to third-party activities.

Mandatory Activities represented 14% (€797 million) whereas 56% was on Optional Programmes (€3067 million), 2% on activities associated to the General Budget (€107 million), 28% on programmes financed by third parties (€1521 million) and 0.1% on others (€5 million).

The core elements of Mandatory Activities are the Scientific Programme and Basic Activities including, among other things, the General Studies Programme, Core Technical Expertise, Common Infrastructure, Technology Research Programme, Earthnet and Long-Term Data Preservation. In 2016, the Scientific Programme cost reached €570 million (10% of the total). Main activities were the development of BepiColombo, Solar Orbiter, JWST, Euclid, CHEOPS and JUICE as well as the preparation of future missions. In addition, the Scientific Programme funded the operations of ten satellites in orbit. The Rosetta spacecraft terminated its mission on 30 September and Philae was located through close-up images captured by the OSIRIS camera.

The execution of programmes including the study, development, launch and operations of satellites as well as the development, exploitation and maintenance of launchers and other space systems is implemented via the Agency's Optional Programmes, in which Member States participate with a voluntary subscription.

In 2016, €1176 million, or 21% of total Agency cost, related to the Launchers Programmes (excluding the Guiana Space Centre). Ariane exploitation activities supported six launches of Ariane 5 including an Ariane 5 ES which injected four Galileo satellites simultaneously in medium Earth orbit on 17 November. Vega Exploitation supported two Vega launches. Ariane 6 Phase-A/B1 development activities concluded with a Programme Implementation Review in September. The Resolution on the continuation of the Ariane 6 and P120C Elements of the development was adopted at the September meeting of Council. Phase-B2 activities started and the prime industrial contract for Phase-C/D/E1 was signed in November. For Vega-C, following the conclusion of Phase-B activities, the reorientation of Vega-C

took place, implementing the changes to the P120C and Vega-C first stage. The Preliminary Requirements Review for Space Rider was completed in December. The Future Launchers Preparatory Programme continued to work on various technologies and integrated demonstrators in accordance with its objectives.

In Earth Observation, €770 million was accrued in 2016, 14% of ESA cost. Sentinel-3A and Sentinel-1B were launched on 16 February and 25 April, respectively. Costs in 2016 covered the development of the Sentinels under the GMES Space Component Programme, with the rider for €180 million for Jason-CS (GSC-3) signed in March, as well as the meteorology satellites with activities for Meteosat Third Generation and MetOp Second Generation, and the development of EarthCARE and ADM-Aeolus (Biomass Phase-C/D signed in April for €230 million) within the family of Earth Explorer missions.

Cost for Human Spaceflight, Microgravity and Transportation, amounted to €419 million, 8% of total cost, related mainly to Columbus operation and development activities for Orion FM1 and FM2. Following approval of the European Exploration Envelope Programme at the Ministerial Council 2016, ExoMars Rider 2 was signed on 16 December, consolidating the industrial contract for the full development of the ExoMars rover and surface platform mission in 2020.

Robotic Exploration and PRODEX cost of €226 million in 2016 corresponds to 4% of total cost, covering mainly ExoMars activities following the Ministerial Council decision. In Telecommunications and Integrated Applications, representing 6% (€330 million) of total cost, the main focus included Competitiveness and Growth, Partner PPPs, Neosat, IAP and the European Data Relay System (EDRS) programme. EDRS-A was launched on 29 January and operational services for Copernicus started in November.

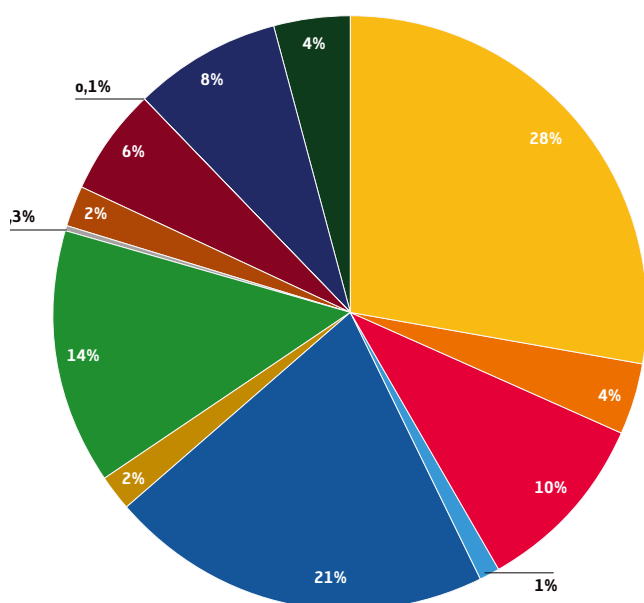
In 2016, the General Study Technology Programme represented 2% of total cost (€100 million) of which €6 million was for Proba-3 development and €94 million applied to R&D activities.

The Navigation programmes represented 1% of ESA's total cost in 2016 (€28 million) covering the European GNSS Evolution Programme.

The Space Situational Awareness Programme cost €17 million (0.3% of total ESA cost). The Space Weather Network and its services have been improved and the SST segment completed the development of a system architecture performance simulator and a visualisation component.

For Third-Party activities, €736 million related to Navigation activities funded by the European Commission (EC) through the Galileo and EGNOS Delegation Agreements and the EGNOS GSA Working Arrangement. The first amendment of the Horizon 2020 Delegation Agreement to cover the 2017 work plan was agreed for an envelope of €230 million to 2023. In December, the working arrangement between ESA and GSA was signed covering the Galileo exploitation phase with a maximum budget envelope of €1400 million to 2021. Activities for Full Operational Capability Deployment Completion (FOC DC) focused on launches of eight satellites as well as on the preparation of the FOC

satellite operations and ground system verification activities. In December, the signature of the FOC DC third amendment of the EC Delegation Agreement took place with a maximum budget envelope of €2,485 million until 2021. In the Earth Observation area, Third-Party activities amounted to €718 million, mainly related to cooperation with the EC on Copernicus MFF (€603 million). The remaining cost is linked to the cooperation with Eumetsat on MetOp and Meteosat Third Generation.



Cost (€ million)

Income

ESA Mandatory Activities (including the General Budget and Scientific Programme) and activities associated with the General Budget are financed by Member and Participating States' contributions. The ESA Convention stipulates that the Agency's Mandatory Programmes are financed using a contribution scale based on the national income of the Member States over the previous three years.

ESA Optional Programmes are financed by Member and Participating States' contributions, and for some programmes, also by other entities, including the European Union and

Eumetsat. Participating States declare a voluntary subscription on a multi-annual basis.

Third Party Programmes, managed by ESA, are fully financed by third parties. In 2016, they represented 29% of total ESA income.

	2016	
	Mandatory Activities (%)	Optional Programmes (%)
Austria	2.21%	1.16%
Belgium	2.74%	5.90%
Czech Republic	1.01%	0.37%
Denmark	1.78%	0.55%
Estonia	0.11%	0.06%
Finland	1.38%	0.38%
France	15.16%	25.60%
Germany	20.42%	25.55%
Greece	1.46%	0.01%
Hungary	0.63%	0.05%
Ireland	1.04%	0.61%
Italy	11.03%	16.26%
Luxembourg	0.19%	0.73%
Netherlands	4.53%	2.35%
Norway	2.81%	1.32%
Poland	2.74%	0.30%
Portugal	1.15%	0.24%
Romania	0.90%	0.68%
Spain	7.81%	3.17%
Sweden	2.82%	1.83%
Switzerland	3.87%	4.09%
United Kingdom	14.21%	8.46%
Total MS	100%	99.68%
Canada	4.38%	0.32%
Total Cooperating	4.38%	0.32%
Slovenia	0.18%	
Total Associate MS	0.18%	
ESA	104.56%	100%

Contributions from Member and Participating States in 2016 to Mandatory Activities, Optional Programmes and associated budgets

Long-Term Planning and Performance Management

The implementation of the ESA Long-Term Plan (LTP) 2015–24 was assessed with respect to the achievements in 2015 and of the progress made on the achievement of actions stemming from the programmatic decisions at the Ministerial Council meetings in 2012 and 2014, as well as on the implementation of programmes and activities in line with sectoral priorities and objectives.

Focus was also placed on the status of implementation of approved optional programmes, presented to the Council Working Group along with the status of the preparation of

programme proposals, as well as the elements in support of the DG's proposal for the mandatory activities. All these elements contributed throughout the year to the preparation of the LTP 2017–26, which was issued and discussed for the first time during Council in October, before being refined and presented as a background document at Ministerial Council 2016.

Consultation meetings with delegations were organised to allow the Executive to estimate possible subscriptions to the programme proposals being put forward to finalise the Ministerial Council Package for mandatory and optional activities.

In parallel with the preparation of Ministerial Council 2016, the strategic guidelines of the Agency were updated, based on the DG's Vision and on the joint European goals for space.

On the basis of political and programmatic decisions taken by Member States and the actual subscriptions collected, the draft LTP was consolidated and submitted to the December meeting of Council according to its nominal planning cycle linked to the preparation of 2017 budgets.

Following the decisions taken at Ministerial Council 2016 on the evolution of the Agency, actions and initiatives aimed at aligning the organisation with Space 4.0 for a 'United Space in Europe' were initiated. The first ESA Corporate Development Plan including short-term measures as well as longer-term actions in selected areas of corporate activities is under preparation.

Performance Management and KPIs

The annual report on the performance of the Agency in 2015 was presented to the AFC at its 1 June meeting in Stockholm and to Council at its 15–16 June meeting. An outlook on the evolution of the strategy map was provided, based on internal reflection and external consultation (delegations and important stakeholders in the European space community) in addition to the usual updated measurements of Agency-level key performance indicators (KPIs), assessed with respect to the targets set for achieving ESA strategic objectives as defined in the Strategy Map of 2012, and reports on the progress of programmes and activities.

Each year after the summer break, an interim report is provided to the Director General and the Executive Board with the aim of triggering possible corrective actions to improve performance, if needed.

Driven by the joint European goals for space, the ESA Strategy Map has been under revision in order to align the measurement of the Agency's performance to the evolving strategic objectives. A status report on the progress achieved in defining a new Strategy Map and identifying candidate indicators to measure the Agency's performance was presented for consultation to the AFC at its meeting at the end of September. Based on the guidance received from delegations and on the actual outcome of Ministerial Council 2016, the consolidation of the strategic objectives of the Agency and the revision of the Strategy Map together with new key performance indicators, are ongoing in the framework of a wider process of corporate development.

Corporate Risk Management

The Agency-level Risk Management process was implemented during 2016 on the basis of the relevant policy and plans. Two meetings of the Network of Risk Coordinators took place as planned in April and October to update and consolidate the Agency Risk Register, with progress being made on mitigation actions as well as the general status of existing risks. The outcome of the assessments was included in the Annual Report, presented to Council at its June meeting and included in the consolidated LTP provided to Member States at the Council meeting in December.

At the beginning of September, the 4th edition of ERMINE, the Enterprise Risk Management International Network jointly founded by ESA, CERN and the CEA (French Alternative Energies and Atomic Energy Commission) took place at the German Aerospace Center in Cologne and gathered Chief Risk Officers from more than 30 private and intergovernmental organisations, all in the high-tech area, and from all around the world. They held exchanges on best practices in enterprise risk management and created an international group of experts reflecting on new methodologies to enhance risk management practices. European Intergovernmental Scientific Research Organisations, on an initiative led by CERN, met in October for the first EIROforum ad-hoc working group on enterprise risk management.

Legal Services

In 2016, the ESA Council took a number of decisions among which the adoption of:

- an Act in Council on funding for the ExoMars programme activities (ESA/C/CCLVIII/Act.1 (Final));
- a Resolution for establishing a Navigation Innovation and Support Programme (NAVISP) (ESA/C/CCLVIII/Res. 2 (Final));
- a Resolution on the continuation of the Ariane 6 and P120C Elements of the Ariane and Vega development Programme (ESA/C/CCLX/Res. 1 (Final));
- a Resolution on the continuation of the Ariane 6 and P120C Elements of the Ariane and Vega development Programme (ESA/C/CCLXII/Res. 1 (Final));
- a Resolution setting up the ESA Grand Challenge (ESA/C/CCLXIII/Res. 1 (Final)); and
- a Resolution on the reform of the external audit system (ESA/C/CCLXV/Res.2 (Final)).

In addition, at the Ministerial Council on 1–2 December, Council adopted the following four Resolutions:

- Resolution 'Towards Space 4.0 for a United Space in Europe' (ESA/C-M/CCLXIV/Res.1 (Final));
- Resolution on the level of resources for the Agency's mandatory activities 2017–2021 (ESA/C-M/CCLXIV/Res.2 (Final));
- Resolution on the CSG 2017–2021 (ESA/C-M/CCLXIV/Res.3 (Final)); and

- Resolution on ESA programmes: outlook and way forward (ESA/C-M/CCLXIV/Res.4 (Final)).

Furthermore, 16 agreements were concluded with ESA partners (see list in Annex), including an association agreement with Slovenia on 5 July (ESA/LEG/458).

ESA Resolutions adopted by Ministerial Council and ESA Regulations are available on the 'Law at ESA' pages of the ESA website.

External Relations

Relations with Member States

As an intergovernmental organisation, ESA first and foremost pursues the collective interest of its Member States. With 22 Member States in 2016 and Canada, it was key to understand the drivers, stances and aspirations of individual Member States not only to contribute to the Agency's decision-making, but also to ensure alignment and consensus on ESA's future development, thereby strengthening Europe's future potential in space.

To this effect, a number of important steps and initiatives were taken in 2016. The existing practices to coordinate relations with Member States were expanded, as the Director General gave priority to visiting ESA Member States ensuring a thorough exchange with Ministers and Delegations on the programmatic as well as the strategic objectives ahead, paving the way for the 2016 Ministerial Council, held in Lucerne, Switzerland.

The preparations for the Ministerial Council resulted in intense activity in support of decision-making, accompanying a high number of preparatory meetings with the Heads of Delegation, including internal preparation through the work of the Inter-Directorate Working Group. The broader opportunities for Member States to express and discuss their expectations with the Director General, and through working meetings, enabled support in drafting the Director General's Proposal and the Resolution 'Towards Space 4.0 for a United Space in Europe', which was – along with three other Resolutions – adopted in Lucerne.

Finally, a comprehensive database of key facts and figures for each Member State and its involvement in space was further expanded, to ensure continuous and growing support and understanding of the national context of each of the Agency's Member States and their involvement in space activities and policies.

Relationship with EU institutions

There were several events in 2016 aimed at shaping the future relationship between ESA and the European Union. On 30 May, the Dutch Presidency of the EU Council held an Informal Space Ministerial Meeting in The Hague. It provided an opportunity for ESA, its Member States and the EU to coordinate their strategies and policies and to assess the conditions for ESA/European Commission (EC) cooperation on EU Space Programmes. The Member States reaffirmed their strong preference for the EC and ESA to formulate a common approach to developing

a European space strategy. In response, the EC proposed developing a joint EC/ESA joint statement on their visions and goals for the future, as an alternative to a single document for a European space strategy.

In the follow-up to this meeting the ESA Director General met on a bilateral level with the Director-General of the Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs, Lowri Evans, throughout April and June to discuss such a joint statement. These meetings presented an opportunity for them to discuss their respective visions for future cooperation, in the context of ESA's preparation of Ministerial Council in December and the EC's preparation of its Communication on a 'Space Strategy for Europe' released in October.

During ESA's 260th (extraordinary) Council meeting, held on 13 September, Commission Vice-President Maroš Šefčovič addressed the ESA Council delegates. He praised the success of ESA/EU cooperation so far on GNSS and Copernicus, and advocated the need to combine mutual strengths and to clarify and simplify the relationship to increase its efficiency and reach common objectives.



Maroš Šefčovič, Vice-President of the European Commission, delivered an address to the ESA Council, on 13 September

On 26 October, the ESA Director General signed a Joint EU/ESA Statement on the 'shared vision and goals for Europe in space'. In this joint statement, ESA and the EU list a number of common goals and emphasise their intention to reinforce their cooperation in the future. On the same day, the EC released its 'Space Strategy for Europe', detailing its proposal for future EU space activities and particularly underlining what is linked to the space economy and to 'new space'.

On 29 November, Commissioner Bieńkowska presented the EC's 'Space Strategy' to the Ministers of the EU Member States during a meeting of the EU Council in its 'Competitiveness' formation. On this occasion the ESA Director General delivered a speech to the EU Ministers welcoming the EC's Communication and explaining ESA's priorities in preparing the ESA Ministerial Council in Lucerne. On the same day the ESA Director General also spoke at the Committee for Industry, Research and Energy (ITRE) at the European Parliament, a legislative committee, which follows up on the commission's space strategy and all legislative proposals arising from it.

On 1 and 2 December, ESA held its Ministerial Council in Lucerne. On the morning of 2 December, Commissioner

Bieńkowska addressed the ministers explaining the rationale of the EC's Space Strategy and stating that the ESA/EU Joint Statement was the first political step towards enhanced cooperation. Finally, on 15 December, the EC, GSA and ESA signed a delegation agreement on the exploitation phase of the Galileo programme. This delegation agreement involved a major revision of the project management plan for deployment and exploitation. The same day, the EC formally announced the start of Galileo Initial Services, the first step towards full operational capability for Galileo.



The 'Joint Statement on Shared Vision and Goals for the Future of European Space' was signed by ESA Director General Jan Wörner and European Commissioner Elżbieta Bieńkowska in Brussels on 26 October

Relationship with EU Member States not Members of ESA

Representatives of the eight EU Member States that are not Members of ESA attended the ESA Ministerial Council in December in Lucerne. The Director General pursued the consultation meetings with those countries in February and June to take stock of the current cooperation status with the Agency and to discuss new opportunities for cooperation.

Slovenia became an Associate State on 16 November, following the signature of Association Agreement on 5 July and the completion of its ratification.

A European Cooperating State (ECS) Agreement with Cyprus was signed on 6 July in Nicosia. Following the signature of the ECS Agreements with Bulgaria and Slovakia in 2015, their respective PECS Charters were both signed in Paris on 4 February thereby making it possible to start implementation of the first contracts.

The Cooperation Agreement with Malta was extended for an additional five years in December.

Two ESA radar courses were organised in the course of the year: the first in Sofia, Bulgaria, and the second in Cēsis, Latvia.

A Space Day was organised in Bulgaria and Slovenia to provide information on European space activities, including interactions with industry.

International relations

United States

Europe celebrated 15 years of presence on the ISS, with ESA astronaut Umberto Guidoni having boarded the station on 23 April 2001. Development of the European Service Module for the NASA Orion capsule continued.

Discussions on possible ESA contributions to the next NASA Mars orbiter mission have begun. Work on current ESA/NASA cooperative space science projects proceeded, with a particular focus on the James Webb Space Telescope and Solar Orbiter. ESA and NASA have agreed to investigate a potential NASA contribution to Athena, the selected ESA 'large-class' X-ray astronomy mission.

The 7th ESA/NASA Joint Programme Planning Group (JPPG) Earth Science and Observation meeting took place at ESTEC on 22 and 23 June. Achievements since the last meeting were presented to the recently appointed ESA Director of Earth Observation and the NASA Associate Administrator for Science. Following an exchange of status and programmes, the meeting received reports from three working groups, and recognised that JPPG has fostered many collaborations that have resulted in benefits to both agencies. The JPPG requested from the working groups that they devise a method to assess the value of this cooperation.

The 'operating arrangements' were finalised on the implementation of Sentinel data access and data exchange with NASA, NOAA, and USGS and the agreement with Eumetsat, NASA and NOAA related to cooperation on the Sentinel-6/Jason CS mission.

Discussions on possible cooperation with NASA and NOAA in the area of space weather were initiated.

Russia

Collaboration activities between ESA and Russia resulted in the completion by the end of 2015 of all tests and development work for the ExoMars 2016 mission and the start of the launch campaign in Baikonur. The ExoMars 2016 mission was launched on 14 March.

Joint work continued on design and development of the ExoMars 2020 mission with the System Preliminary Design Review for the entire mission completed. Within a joint Announcement of Opportunity additional science payload for the surface platform was selected.

Cooperation between ESA and Roscosmos in the area of lunar exploration continued and was given a formal basis with an exchange of letters. The Sentinel-3A launch campaign started in Plesetsk in November 2015 and was followed up by launch of the satellite by Rockot in February.

International Space Station cooperation with Roscosmos continued with British ESA astronaut Tim Peake's six-month Principia mission completed in June 2016. In November 2016, French ESA astronaut Thomas Pesquet began his long-duration ISS Proxima mission as a flight engineer for Expeditions 50 and 51.



The 2016 Dragon 3 Final Results and Dragon 4 Kick-off Symposium was held in Wuhan, China

A programme of joint scientific experiments on board the International Space Station was implemented and included the joint ESA/Roscosmos experiments Plasma Kristall-4, Immuno-2, SODI DCMIX, Electromagnetic Levitator, MARES and EDOS.

Soyuz at CSG cooperation continued with the launches of two Soyuz-ST vehicles.

China

ESA, together with the Chinese Academy of Sciences (CAS), after a peer review, selected the SMILE mission. The two sides are studying the implementation of this collaborative project between ESA and China. The SMILE mission will give European and Chinese scientists scientific data about the interaction between solar wind and Earth's magnetosphere.

The Dragon programme with the National Remote Sensing Centre of China from the Ministry of Science and Technology continued to yield high quality scientific data. Dragon-4 cooperation began in 2016 and is focusing on science and applications development in China using data from ESA and Chinese Earth observation missions. ESA and the China Manned Space Agency are studying the possibility of flying a European astronaut on the Chinese space station in the 2022–23 timeframe.

Japan

The Director General met the JAXA President on 25 September on the margins of the 67th IAC. This was an opportunity for the two heads of agency to express a general wish to expand cooperation beyond the current joint missions, BepiColombo and EarthCare, and to call for a related review of cooperation opportunities.

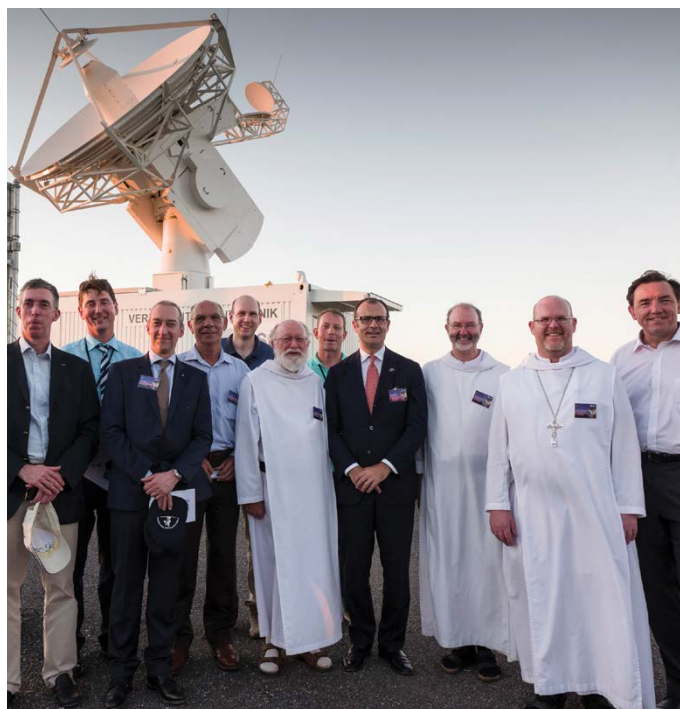
India

The Director General met the Chairman of ISRO at ESA headquarters on 6 October to reenergise bilateral cooperation through the creation of a dedicated working group tasked with studying possible activities of joint interest. The current ESA/ISRO agreement was extended by another period of five years, up to 2022.

The Director General participated in the dinner of the heads of space agencies in New Delhi in April, which endorsed the New Delhi Declaration on Climate Change.

Australia

A delegation from the European Parliament visited Perth, coinciding with the inauguration of a new TT&C antenna in New Norcia with the participation of the EU Ambassador in Australia. A Technical Operating Arrangement between ESA and Geoscience Australia was signed in 2016 regarding access to Sentinels Earth observation data by Australia. It followed the signature in 2015 of a political Agreement between the EU and Australia covering access to Copernicus data.



Inauguration of the new 'NN02' acquisition antenna at ESA's deep-space tracking station, New Norcia, Western Australia, 11 February

Argentina

Focus was placed on securing, in cooperation with ESA's Argentinian counterpart CONAE, continued operations of the Agency deep space antenna in Malargüe, Argentina and progressing in other ESA-CONAE bilateral cooperation, in particular in Earth observation. Argentina's Ministry of Science, Technology and Productive Innovation, CONAE and ESA organised a Space Day on 26 April in Buenos Aires to present

ESA/Argentina cooperation to more than 180 participants from the Argentinian scientific community.

Support to Global Challenges on Earth

In order to support those who can benefit from space projects, applications or services, a cross-cutting approach under major themes or geographical areas has been developed for a number of years across Directorates and has gained momentum. In particular, the adoption by the United Nations in September 2015 of its 17 Sustainable Development Goals (SDGs) has been followed up at ESA by the development of a specific 'Space for Earth' (www.esa.int/spaceforearth) page on Sustainable Development and a draft catalogue of ESA activities supporting UN SDGs (accessible through www.esa.int/SDG).

It was accompanied by presentations to ESA Member States of the concept of a 'Coordination Platform' that would use as seed inputs the ESA catalogue, and be complemented by equivalent catalogues from other space agencies or service providers, in order to facilitate the identification by any ground user of the space projects or services that could help support the achievement of an SDG. ESA activities can indeed support all of the 17 Goals and a full catalogue is under development. Discussions have also started with the United Nations Office of Outer Space Affairs (UNOOSA), which plans to produce a Solutions Compendium with the same aim. The ESA catalogue would provide the 'technical' initial input to the compendium. This activity is being performed in the framework of the preparation of the Unispace+50 conference in June 2018.

An analysis of the partnerships signed by ESA with non-space entities was also started, with the aim of developing them further, and identifying new partnerships, supporting better knowledge and use of ESA activities by those tackling global challenges.

In parallel to the approach by theme mainly based on the SDGs, an approach by geographical area has been developed focusing on the interests, priorities and concerns expressed by Member States. Support to the Arctic region in 2016 took the form of the setting up of an internal Arctic Task Force, followed at the end of the year by an external ESA Space and Arctic Task Force with representatives from 15 states. Activities in preparation are aimed at supporting both a better knowledge of the Arctic environment and the economic development of the region. Similar support has been started and will develop further for the Atlantic area, Alps and Africa.

Communication

The focus of activities in 2016 was on improving awareness and informing our main target audiences – the general public, political decision-makers and opinion formers/media – about the activities of ESA and its Member States, and how those activities benefit society, in promoting industrial competitiveness and inspiring the younger generations.

While the most closely followed missions were Rosetta, ExoMars, Principia and Proxima, the long list of ESA mission milestones provided excellent opportunities to convey



ESA astronaut Timothy Peake with one of his educational activities, Space Seeds, on the International Space Station

messages and to reach ever-larger audiences. This included the launches of EDRS-A, two Sentinels and six Galileo satellites, Rosetta's 'grand finale' in September, as well as the launch of ExoMars 2016 in February and its arrival at Mars in October.

A particular highlight was the campaign for astronaut Tim Peake's mission, which created immense interest through its media and education activities. This was thanks to the cooperation with the UK Space Agency and other partners from the education, charity, music, sports and publishing industries.



ESA astronaut Timothy Peake drew huge crowds during media events at Farnborough Airshow

The number of followers of Tim's Facebook page grew from fewer than 20 000 before his launch in December 2015, to almost 700 000 by the end of his mission, representing the highest number of Facebook followers of any ESA astronaut to that point. The same growth was observed in his Twitter following, and Tim is still the third most followed astronaut on Twitter globally.

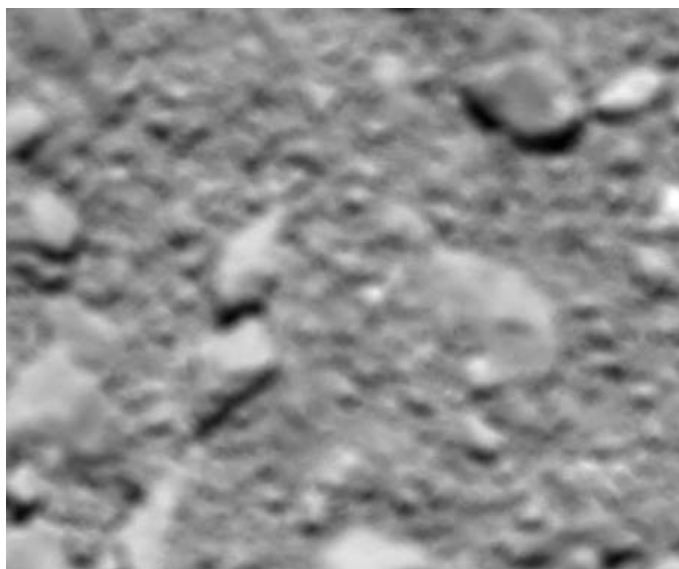


Vangelis's first studio album in over 15 years was dedicated to the ESA Rosetta mission and promoted ESA to a wider audience (ESA/Universal/Salvador Design)

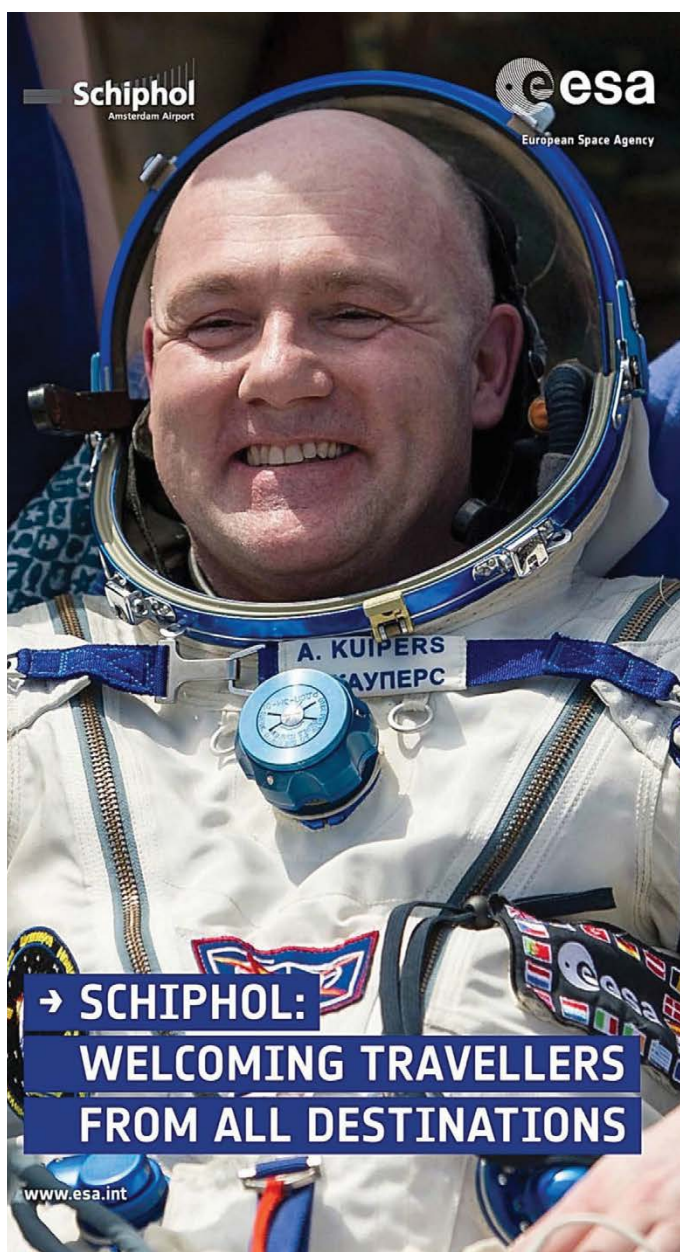
The final hours of Rosetta's descent and landing on Comet 67P were relayed live from the main control room at ESOC in a dedicated video broadcast to the general public and broadcasters worldwide. Drawing on the descent images, the broadcast generated huge coverage on Facebook, Twitter and Instagram. ESA's Rosetta mission Twitter account reached over 400 000 followers. The media impact of the mission, from the wake-up in January 2014 to the end of mission in September, was estimated by Kantar Media at 4.9 billion contacts.

After Rosetta, the ExoMars landing was ESA's second largest communication effort including the specific response to the crisis situation. While media reports covered mainly the cause of the non-nominal landing and its repercussions for the 2020 rover mission, ESA Communications moved its focus to the Trace Gas Orbiter, its first science and the preparations for aerobraking. A total of 673 000 people watched the livestream coverage and 64 000 watched the press conference after landing. The landing generated 61 000 tweets. The global reach amounts to 660 million contacts via traditional and social media.

The year was busy for Galileo, with several opportunities to maintain and raise public interest in Europe's satellite navigation system. After a standard launch on Soyuz at the beginning of the year, in November an Ariane 5 rocket was scheduled to place four Galileo satellites into orbit: the first-



Rosetta's last image of Comet 67P/Churyumov-Gerasimenko, taken shortly before impact, generated wide interest on social media



One of a series of posters for the Schiphol Airport collaboration (ESA/Schiphol Airport)

ever quadruple launch made possible by a new payload dispenser capable of releasing four identical satellites into orbit in one go. This was the eighth Galileo launch, taking the number of satellites in space to 18.

Finally, in mid-December, the European Commission, as owner of the system, formally announced the start of Galileo Initial Services, the first step towards full operational capability.

ESA Communications has also continued to maintain and generate partnerships with space and non-space players, such as major news agencies media outlets (BBC, Euronews, Channel 4, RAI, ARD, El Pais and so on), the ECSITE network of science centres and global movements such as International Asteroid Day.

New initiatives of note were the #SpaceRocks social media activity during Tim Peake's mission (involving more than 50 individual contacts in the UK music industry), the release of the 'Rosetta' album by world-famous musician Vangelis (featuring music composed specifically for ESA in the first collaboration with a major music company) and the joint poster promotion at Amsterdam Schiphol Airport.

Education and Knowledge Management

In close collaboration with numerous partners in the Member States, the ESA–ESERO approach to formal school education in Europe continued to prove its effectiveness. The impact of using space as a learning and teaching context in the classroom was evidenced by teachers in several countries reporting an increase in knowledge of the science and technology behind the units they teach, as well as registering higher interest and attainment on the part of their pupils. About 10 000 primary/secondary teachers attended training sessions, potentially reaching out to about 300 000 pupils.



Winners of 2016 ESA/EISC Space for Sustainability Award in Bucharest in October

At tertiary education level, the development of the two new ESA Redu-based facilities for university education (Training and Learning Centre and European CubeSat Lab) have played a major role in enabling the widening of the scope of the whole ESA tertiary education programme, allowing the definition of the new ESA Academy programme.

Officially announced in March, this new ESA overarching programme for university education encompasses the long and well-established Hands-on Space Programme and the new Training and Learning Programme aimed at facilitating a well-structured transfer of ESA expertise and best practices to students; as well as access to state-of-the-art technical facilities. No fewer than 1500 students benefited from the various opportunities offered by the ESA Academy Programme; including three university student teams from Italy, Belgium and Denmark who launched their CubeSats on the same Soyuz as Sentinel-1B in April from Europe's Spaceport in Kourou, French Guiana.

Sustainable Development and Corporate Responsibility

ESA began institutionally to commit towards sustainable development and report on its activities in this area almost 10 years ago. This voluntary approach (contrary to the mandatory one for most companies) was both of a top-down and bottom-up nature. It corresponded to a political will to build up a holistic and harmonised approach to sustainability for the space sector concerning both core business and support activities.

It was also in response to a strong desire on the part of ESA staff to see their Agency become more aligned with its fundamental causes and missions (delivering applications and services for the benefit of society, planetary security and health) in its decision-making processes and in its everyday operations with sustainable development or corporate social responsibility values and requirements. Staff expressed these considerations in a call for ideas launched in 2007 and a citizens' questionnaire on sustainable development also issued in 2007.

ESA had committed to be exemplary as a space agency and as an organisation and had the ambition to be a model agency for clean, responsible and sustainable space activities. ESA has consequently developed a sustainable development framework to meet legal and regulatory requirements in its core business and support activities, to meet stakeholders' expectations of a fair and transparent reporting of its activities, to guarantee accountability for sound, efficient use of public money and to ensure ESA responsibility.

ESA is now a leading space agency that operates a comprehensive and mature sustainable development policy based on a cross-cutting approach that comprises actions both in outer space and on the ground. The Agency paved the way for other space actors to commit towards sustainable development, and shares policy elements and best practices on a regular basis through various channels.

Lastly, seen through the prism of society, space programmes and applications support public policy and can also contribute to addressing any of the challenges of the UN SDGs that have now become a central plank of governance worldwide and also even at business level.

ESA's fourth biennial SD report based on 2015–16 data has now been prepared. The purpose of reporting on sustainability activities is a matter of awareness, assessment and transparency.

To be able comprehensively to assess sustainability at ESA, generic and specific key performance indicators (KPIs) with their dedicated management goals are being deployed strategically across the Agency. They structure the reporting scheme based on international standards such as the Global Reporting Initiative (GRI).

As in previous years, the report will be structured around five chapters: governance, space contributing to a sustainable society, managing our environment responsibly, relations with partners and suppliers and human resources and social responsibility.

The publication of the 2015–16 ESA Sustainable Development report now renamed 'Corporate Responsibility and Sustainability' is planned for September 2017 and will be available online at www.esa.int.

In 2016, the 4th Edition of the ESA/EISC Space for Sustainability Award took place. The 'Space for Sustainability Award' targets students and young professionals under 30

with project ideas that will enable the space sector to put sustainable development on a formal footing, raise awareness among the younger generations and professionals, and contribute to furthering the debate on the topic. The award ceremony took place during the EISC plenary in Bucharest on 3 October, where the two best candidates were rewarded and invited to present their project ideas before the parliamentarians. Eric Herrero won first prize for his project idea entitled 'Satellite prediction of illegal wildlife trafficking routes'.

The previous years of the competition gathered 41 participants from 13 different European countries. The winners of the previous editions all had the opportunity to attend a launch from Europe's Spaceport in Kourou and witnessed the Gaia, Galileo 7 and 8, Galileo 11 and 12 and SmallGEO Hispasat 26W-1 launches.

ESA started its cooperation with NASA in the field of sustainability in 2009 and has been co-organiser of an annual international workshop on environment and alternative energy since 2011 (which started at ESTEC) also with the support of the Portuguese Centre for the Prevention of Pollution (C3P).

The workshop was held in October at the NASA Jet Propulsion Laboratory (JPL), in California. The event brought together European and American partners to talk about sustainability, from ESA and NASA but also from industry, startups, academia and other parties. Some global issues that are common to all were discussed (environmental management, critical infrastructure, risk management, resilience factors, cybersecurity, dialogue with stakeholders, and so on), but also some more space-related issues such as materials and processes (with a focus on the REACH European regulation).

Lastly, a special project was developed by the Coordination Office on Sustainable Development in relation to an innovative stakeholders' dialogue. On 10 September, the Citizens' Debate on Space for Europe took place in each of the 22 ESA Member States. This one-day event was the first consultation ever organised between a space agency and its main stakeholders: citizens and taxpayers. More than 2000 participants attended and debated space issues in the course of a day-long meeting. ESA organised the debate to involve new participants to develop its future orientation and nurture its strategy. It is crucial to the Agency to involve more actors, especially from the general public, in choices that will respond to societal challenges, put space at their service and affect coming generations through space innovation, exploration and international cooperation.

Security Office

Of the eight security inspections performed at ESA establishments and stations in 2015, three were performed by an external contractor, selected to provide a fresh, unbiased view of the status of security at these sites (three more such external inspections will be performed in 2016). Security issues at corporate level were addressed with bilateral meetings with the departments responsible, namely Facility Management, Information Technology, and Procurement; this



Winners of 2016 ESA/EISC Space for Sustainability Award in Bucharest in October

approach has been consolidated, and will also be extended to Human Resources in 2016.

Reaccreditations of the classified installations related to the Galileo Project were performed at ESTEC (satellite-testing facilities, the PRS laboratory, Project office premises in the T building), at Redu and at ESOC. The project was supported by participation in several security boards and in the accreditation panel.

The Launchers Directorate was supported in the detailed definition of the Security Classification Guide. This annex to the

Programme Security Instructions already adopted in 2015 specifies the level of classification of all components to be formally protected.

The security measures implemented across the Agency enable ESA to undertake security-related programmes. A new project, 'Cyber Range', was started to develop space mission security in which future mission scenarios for different categories can be simulated and tested with the aim of ensuring that security risks, in particular cyber-related, are properly addressed before it is too late.

Notes to the Financial Statements

As required by the ESA Financial Regulations, the Agency's annual Financial Statements are prepared in accordance with International Public Sector Accounting Standards (IPSAS) since 2010.

The Financial Information presented herewith was extracted from the audited ESA Financial Statements for the financial year 2016. This Financial Information consists of the following extracted components:

- Statement of financial position;
- Statement of financial performance;
- Statement of changes in net assets/equity;
- Cash flow statement.

A transition period was put in place during the years 2010–15 for the progressive and full recognition of the Agency's fixed assets as provided for under IPSAS 17 for property, plant and equipment. This period ended on 31 December 2015 and, thanks to the improvements made during the transition period, resulted in a clear and full certification of the audited ESA Financial Statements.

In 2016 special attention was paid to the consolidation of the earlier achievements, in particular in the area of fixed assets, with the reinforcement of internal controls over the accounting and financial reporting process.

Some items, such as fixed assets, subject to first IPSAS recognition, are presented in these statements with only the counterpart in the reserves, but without impact on the statement of financial performance.

At the end of 2016, final fixed assets amount to €3681.4 million and assets under construction to €3587.8 million net.

The Net Deficit of 2016 amounts to €269 million, after capitalisation of Assets Under Construction and work-in-

progress, as compared to a deficit of €109.8 million in 2015 (restated).

The increase of the 2016 deficit of around €159 million compared to 2015 is mainly due to the increase of operating expenses of around €922 million not fully compensated by the increase of operating revenues (€761 million).

The 2016 net deficit breaks down into::

- €34.1 million net surplus for ESA (€319.2 million in 2015)
- €303.4 million net deficit for Pensions (€431.1 million in 2015)
- €0.3 million net surplus for Social Security (€2.1 million in 2015)

The Financial Information presented hereinafter, was (as stated above) extracted from the ESA Financial Statements 2016. ESA Financial Statements 2016 also include a summary of significant accounting policy and notes to the Financial Statements, and were audited by the Audit Commission of the European Space Agency. The Audit Commission conducted its audit in accordance with the International Standards on Auditing (ISA). As required by the Financial Regulations, this audit included also the aspect of regularity.

In the Audit Commission's opinion, the 2016 audited Financial Statements give a true and fair view of the financial position of the European Space Agency as at 31 December 2016, and of its financial performance and its cash flows for the 12 month period ended 31 December 2016, in accordance with International Public Sector Accounting Standards (IPSAS).

Concerning the aspect of regularity, the Audit Commission reported that nothing came to its attention, which would indicate that the transactions of ESA have not been made, in all significant respects, in accordance with the Regulations of the European Space Agency.

The audited Financial Statements and the Independent Auditor's report were submitted to the Council in accordance with the ESA Financial Regulations.

STATEMENT OF FINANCIAL POSITION

	2016 Million €	2015 restated Million €
TOTAL ASSETS	10 926.5	10 983.5
Current assets	2367.1	2203.9
Cash and cash equivalents	847.2	963.5
Unrestricted cash and cash equivalents	192.4	298.2
Restricted cash and cash equivalents	654.8	665.3
Receivable from non-exchange transactions	496.1	348.1
Outstanding contributions Member/Participating States	432.4	285.4
Outstanding contributions EC	0.4	0.6
Member States reimbursable taxes	63.3	62.1
Receivable from exchange transactions	669.9	674.5
Other customers	39.2	12.1
Prepayments and other amounts receivable	630.7	662.4
Inventories	353.9	217.8
Non-current assets	8559.4	8779.6
Long term receivables from non-exchange transactions	2.8	2.7
Long term receivables from exchange transactions	578.8	981.0
Financial assets	603.6	539.3
Land & buildings	1074.3	1125.0
Technical plant and equipment	576.4	485.9
Spacecraft and space laboratories	1603.1	1546.2
Administrative equipment and vehicles	293.7	289.0
Intangible fixed assets	133.9	138.6
Assets under construction	3692.8	3671.9
TOTAL LIABILITIES	6865.1	6658.3
Current liabilities	3089.8	2854.4
Payables under exchange transactions	1902.8	1883.4
Current payables	776.5	738.4
Pre-financing from exchange transactions	1063.1	1117.0
Deferred income	63.2	28.0
Contributions and other payables	1187.0	971.0
Prepaid contributions and other amounts payable to Member/Participating States	239.6	125.2
Pre-financing from non-exchange transactions	9.8	37.2
Other payables	142.6	168.2
Other current financial liabilities	250.0	140.0
Employee benefits	216.0	198.2
Provisions	329.0	302.2
Non-current liabilities	3775.3	3803.9
Employee benefits	3774.6	3428.7
Pre-financing from non-exchange transactions	0.0	8.2
Pre-financing from exchange transactions	0.7	367.0
NET ASSETS	4061.4	4325.2
Reserves	7318.7	6992.1
Amounts to be called from Member States	-2,988.3	-2,557.1
Net Surplus/(Deficit) for the period	-269.0	-109.8
TOTAL NET ASSETS	4061.4	4325.2

STATEMENT OF FINANCIAL PERFORMANCE

	2016 Million €	2015 restated Million €
Operating revenues	4869.0	4108.1
Contributions from Member/Participating States	3605.4	3354.7
Other contributions and grants	149.2	191.9
Revenues from exchange transactions	862.7	327.9
Social security	17.9	16.7
Pension schemes	24.0	22.2
Other operating income	209.8	194.7
Operating expenses	5159.6	4237.0
Purchases	4033.6	2596.0
External services	896.8	942.7
Staff expenses	304.4	297.7
Social security benefits	32.5	29.5
Pension and other post-employment benefits	451.2	568.5
Other operating expenses	210.2	201.1
Expenses from non-exchange transactions	102.5	107.3
Expenses transferred to assets	-1880.0	-1012.3
Delegated procurement expenses transferred to amounts receivable	-651.7	-415.3
Expenses transferred (to)/from work in progress	-136.1	221.6
Expenses transferred to fixed assets under construction	-1092.2	-818.6
Depreciation	694.1	716.9
Write-downs inventories and fixed assets	290.0	-216.6
Provisions	24.3	6.2
NET SURPLUS / (DEFICIT) FROM OPERATING ACTIVITIES	-290.6	-128.9
Financial activities	21.6	19.1
Financial revenue	23.1	22.4
Financial expenses	1.5	3.3
NET SURPLUS / (DEFICIT) FROM ORDINARY ACTIVITIES	-269.0	-109.8
Extraordinary items	0.0	0.0
NET SURPLUS / (DEFICIT) FOR THE PERIOD	-269.0	-109.8

STATEMENT OF CHANGES IN NET ASSETS/EQUITY

		1 January, 2015 Million €	Errors – 2015 opening balance impact Million €	1 January, 2015 Restated Million €	Movements 2015 Million €
RESERVES	A	8701.2	-19.0	8682.2	-1690.1
Member States capital and reserves		4503.6	-42.4	4461.2	-985.1
Revaluation differences		4454.8	-42.4	4412.4	-970.3
Currency Exchange Gains Reserve		5.3		5.3	-1.7
Sundry reserves		5.1		5.1	-5.0
Other reserves		38.4		38.4	-8.1
Social security reserve		22.6		22.6	2.5
Buffer Fund fair value reserve		15.8		15.8	-10.6
Accumulated Surplus		4197.6	23.4	4221.0	-705.0
AMOUNTS TO BE CALLED FROM MEMBER STATES	B		0.1	-2403.9	-153.2
Member States pension scheme receivable		-2404.0		-2403.9	-153.2
Accumulated Deficit		-2403.9	0.1		
Net surplus / (deficit) for the period	C	-0.1	-27.4	-1856.0	1733.6
TOTAL NET ASSETS	A+B+C	-1828.6	-46.3	4422.3	-109.7

		Errors – 2015 impact Million €	Year 2015 restated Million €	Allocation of 2015 surplus/ (deficit) Million €	Movements 2016 Million €	Year 2016 Million €
RESERVES	A	0.0	6992.1	321.4	5.2	7318.7
Member States capital and reserves		-3.4	3472.7	-618.8	5.5	2859.4
Revaluation differences		-3.4	3438.7	-620.6	0.8	2818.9
Currency Exchange Gains Reserve			3.6	-0.3		3.3
Sundry Reserves			0.1			0.1
Other Reserves			30.3	2.1	4.7	37.1
Social security reserve			25.1	2.1		27.2
Buffer Fund fair value reserve			5.2		4.7	9.9
Accumulated surplus		3.4	3519.4	940.2	-0.3	4459.3
AMOUNTS TO BE CALLED FROM MEMBER STATES	B		-2557.1	-431.2	0.0	-2988.3
Member States pension scheme receivable			-2557.1	-431.2		-2988.3
Accumulated deficit						
Net surplus / (deficit) for the period	C	12.6	-109.8	109.8	-269	-269.0
TOTAL NET ASSETS	A+B+C	12.6	4325.2	0.0	-263.8	4061.4

STATEMENT OF CASH FLOW

	2016 Million €	2015 restated Million €
Net cash flows	-121.0	235.1
Net cash flows from operating activities	930.7	970.1
Surplus/(deficit) from ordinary activities	-269.0	-109.8
Non-cash movements	1008.4	506.5
Depreciation and impairment	694.1	716.9
Provisions	24.3	6.2
Write-downs inventories and fixed assets	290.0	-216.6
Decrease (Increase) in deferrals/accruals of past or future operating cash receipts or payments	191.3	573.4
Decrease/(increase) in inventories	-136.1	221.6
Decrease/(increase) in receivables from Member States	-148.3	124.0
Decrease/(increase) in other receivables	375.2	-191.9
Decrease/(increase) in other current assets	31.7	-277.0
Increase/(decrease) in payable to Member/Participating States	113.2	-80.1
Increase/(decrease) in pre-financing	-455.8	341.6
Increase/(decrease) in other payables	47.7	-53.5
Increase/(decrease) in provisions for employee benefits	363.7	488.7
Net cash flows from investing activities	-1161.7	-875.0
Acquisition of fixed assets	-1097.4	-823.1
(Increase)/decrease in financial investments (portfolio)	-64.3	-51.9
Net cash flows from financing activities	110.0	140.0
Increase/(decrease) in financial debt	110.0	140.0
Net increase/(decrease) in cash and cash equivalents	-121.0	235.1
Cash and cash equivalents at the beginning of the year	963.5	739.0
Effect of changes in fair value of cash and cash equivalents	-4.7	10.6
Cash and cash equivalents at year-end	847.2	963.5

Agreements signed in 2016

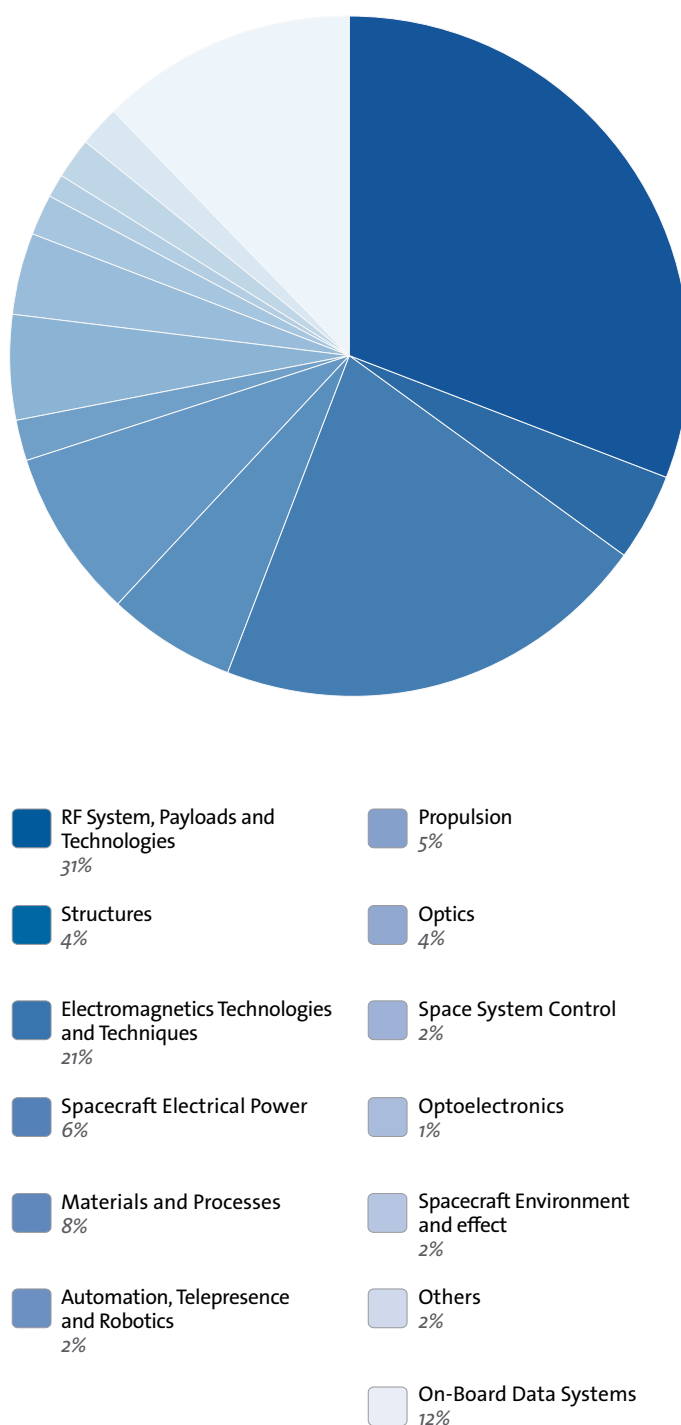
- The fifth extension of the Agreement between the Government of the Federative Republic of Brazil and the European Space Agency on the setting up and use of Tracking and Telemetry Facilities on Brazilian Territory was formalised through an exchange of letters between the Ambassador of Brazil in France, Mr Jose Mauricio Bustani, on 17 September 2014 and JohannDietrich Wörner, the Director General of ESA, on 21 January 2016. **(LEG/168, add. 6)**
- The PECS Charter between the European Space Agency and the Government of the Republic of Bulgaria was signed in Paris on 4 February 2016 by the Head of the Political Cabinet of the Minister of Economy for the Government of the Republic of Bulgaria, Mr Philip Kirev, and the ESA Director General, Mr JohannDietrich Wörner. **(LEG/446, add. 1)**
- The PECS Charter between the European Space Agency and the Slovak Republic was signed in Paris on 4 February 2016 by Marek Eštok for the Slovak Republic and by the ESA Director General, Mr Johann-Dietrich Wörner. **(LEG/443, add.1)**
- The Extension of the ESA-JAXA Agreement concerning cooperation regarding the Greenhouse Gases Observing Satellite (GOSAT), was signed in Japan on 27 March 2015, and in Frascati on 16 March 2016 by Dr Volker Liebig, ESA's Director of Earth Observation, and Mr Kazuo Tachi, JAXA Director Program Management & Integration Department. **(LEG/455)**
- The Arrangement between The Agenzia Spaziale Italiana (ASI) and the European Space Agency with regard to cooperation on the functioning of the European Astronaut Centre of the European Space Agency, was signed in Paris, on 15 June 2016 by Mr Johann-Dietrich Wörner, ESA's Director General and Mr Roberto Battiston, ASI President. **(LEG/456)**
- The Association Agreement between the European Space Agency (ESA) and the Government of the Republic of Slovenia which was unanimously approved by Council on 14 June 2016 (ESA/CR(2016)4), was signed in Paris on 5 July 2016 by the, Slovenian Minister of Economic Development and Technology, Mr Zdravko Počivalšek, and by the ESA Director General, Mr Johann-Dietrich Wörner. **(LEG/458)**
- The European Cooperating State Agreement (ECS) between the European Space Agency and the Government of the Republic of Cyprus which was unanimously approved by Council on 15 June 2016, was signed in Nicosia on 6 July 2016 by the Cypriot Permanent Secretary to the Ministry of Transport, Communications and Works, Mr Alecos Michaelides and Mr Frédéric Nordlund, Head of the ESA External Relations Department. **(LEG/459)**
- The Implementing Arrangement between the European Space Agency (ESA) and the European Defence Agency (EDA) on their cooperation for the Cyber Defence for Space Project, which was approved by Council on 14 December 2016, was signed in Brussels on 2 December 2016 by Mr Jorge Domecq, EDA Chief Executive, and in Paris on 14 December 2016 by the ESA Director General, Mr Johann-Dietrich Wörner. **(LEG/460)**
- The Exchange of Letters between the European Space Agency (ESA) and the National Aeronautics and Space Administration (NASA) concerning the extension of the duration of the MoU on the SOHO Mission was signed in Paris on 20 December 2016 by Mr Frédéric Nordlund, Head of the External Relations Department for the ESA letter, and in Washington on 20 December 2016 by Ms Karen C. Feldstein, Deputy Associate Administrator for International and Interagency Relations, for the NASA letter in reply. **(LEG/462)**

- The Exchange of Letters between the European Space Agency (ESA) and the National Aeronautics and Space Administration (NASA) concerning the Athena Mission was signed in Paris on 20 December 2016 by Mr Frédéric Nordlund, Head of the External Relations Department for the ESA letter, and in Washington on 21 December 2016 by Mr Gilbert Kirkham, Director of the Science Division for the NASA reply. **(LEG/463)**
- The second extension of the Memorandum of Understanding concerning the Laser Interferometer Space Antenna Mission (LISA Pathfinder), by means of an Exchange of Letters, was approved by Council at its 265th meeting held on 14 December 2016. The Exchange of Letters, signed by Mr Frédéric Nordlund, Head of the External Relations Department for ESA, and Mr Gilbert Kirkham, Director of the Science Division for NASA, was concluded on 20 December 2016 upon reception of NASA's affirmative reply to ESA's Letter. **(LEG/327, add.2)**
- The Exchange of Letters between the European Space Agency (ESA) and the National Aeronautics and Space Administration (NASA) for the amendment to the Memorandum of Understanding concerning the EUCLID Mission was signed in Washington on 19 December 2016 by Mr Gilbert Kirkham, Director of the Science Division for the NASA letter, and on 20 December 2016 by Mr Frédéric Nordlund, Head of the External Relations Department for the ESA letter reply. **(LEG/407, add.1)**
- The Exchange of Letters between the European Space Agency (ESA) and the National Aeronautics and Space Administration (NASA) for the amendment to the Memorandum of Understanding concerning the EUCLID Mission was signed in Washington on 19 December 2016 by Mr Gilbert Kirkham, Director of the Science Division for the NASA letter, and on 20 December 2016 by Mr Frédéric Nordlund, Head of the External Relations Department for the ESA letter reply. **(LEG/407, add.1)**
- The Exchange of Letters between the European Space Agency (ESA) and the Indian Space Research Organisation (ISRO) for the Extension of the Cooperative Agreement signed on 21 December 2016 by Mr Johann-Dietrich Wörner ESA Director General and on 27 December 2016 by Mr A.S. Kiran Kumar, Chairman of ISRO. **(LEG/251, add. 2)**
- Amendment No.1 to the Delegation Agreement between the EU, represented by the EC and ESA on the Implementation of Horizon 2020 – Framework Programme for Research and Innovation, was signed in Brussels on 13 December 2016 and in Paris, on 14 December 2016 by Mr Johann-Dietrich Wörner, ESA's Director General and Ms Lowri Evans, Director-General for the Commission. This Amendment No.1 entered into force on 14 December 2016 and shall remain in force until 31 December 2023. **(LEG/453, add.1)**
- The Working Arrangement between the European GNSS Agency and the European Space Agency was signed in Brussels on 15 December 2016, by Mr Johann-Dietrich Wörner, ESA Director General, and by Mr Carlo des Dorides, Executive Director of the European GNSS Agency. **(LEG/461)**

Patents

At the end of 2016, the portfolio contained 133 inventions protected by 431 patents (granted or patent applications- still under examination by patent offices worldwide).

The inventions cover a wide variety of disciplines represented in ESA. The categorisation of the portfolio by ESA Technology Domain has not changed much since last year. Telecom remains the programme that could benefit the most from the use of the Patents Portfolio.



Invention Disclosures received and patent applications filed in 2016

The Secretariat of the Patents Group receives, on average, between 15 and 20 Invention Disclosures per year and files between five and ten applications per year. The following new patent applications were filed in 2016. They result mostly from decisions taken by the Patents Group in 2015

In 2016, 27 Invention Disclosures were received. The Secretariat initiated the review and/or decision process for all of them in 2016. The largest contribution was from TEC with 11 from TEC-E and seven from TEC-M. A large number of inventions resulted from ESA's R&D contracts. In 2016, 12 out of the 27 disclosures arose from collaboration with industry. This shows that there is a strong involvement of ESA's Technical Officers in trying to find innovative solutions together with industry and this proves the value of ESA's engineers.

ESA Reference	Patent title	Technology Domain	Application number
PAT 702	A directional Coupler and Method of Manufacturing Thereof	RF Systems payloads & technologies	EP16203470.6
PAT 656	A method and apparatus for determining a schedule for contact with a constellation	RF Systems payloads & technologies	1612917.3 (UK)
PAT 595	Double-threshold integrity computation	RF Systems payloads & technologies	PCT/EP2016/o63381
PAT 682	Imaging spectrometer with reflective grating	Optics	EP16191731.5
PAT 675	Interference resilient flexible techniques for payload resource allocation	RF Systems payloads & technologies	PCT/EP2016/o63358
PAT 662	Rechargeable electrochemical lithium ion cell	Propulsion	GR 3749/5-7-2016

ESA patent application filings in 2016

Extensions of Protection decided in 2016

The list below shows the decisions made by the PG with regard to the extension of protection of patent applications.

Patent Number	Title	Decision
PAT 651	Method for creating a constellation of electronic devices for providing optical or radiofrequency operations on a predetermined geographical area, and a system of such a constellation of electronic devices	PCT
PAT 626	Very compact TM ₀₁ mode extractor	EP-US
PAT 630	Manufacturing of a metal component or a metal matrix composite component involving contactless induction of high-frequency vibrations	EP-US
PAT 635	Foil Deployment Mechanism	EP-US
PAT 622	Receiving method and receiver for satellite-based automatic identification systems	US-EP-JP-CA
PAT 624	Hybrid Folded Rectangular Waveguide Filter	EP-US
PAT 616	Method for designing a modulated metasurface antenna structure	EP-US
PAT 658	Interferometric Radio Occultation	EP-US
PAT 579	Digital beam-forming network having a reduced complexity and array antenna comprising the same	US
PAT 625	Lumped element rectangular waveguide filter	US
PAT 634	Imaging Antenna Systems with Compensated Optical Aberrations Based on Unshaped Reflectors	US-EP
PAT 637	Wideband Reflectarray Antenna for Dual Polarization Applications	US-EP

Extension of protection decided in 2016

Patents granted in 2016

2016 was a good year for the Agency as some 30 patents were granted in different countries. Those titles are enforceable in the countries where they were granted, reinforcing the position of ESA and its industry

ESA Reference	Patent title	Country
PAT 578	A Deployable Tensegrity Structure especially for Space Applications	ES, UK
PAT 621	Deployable Antenna Frame	US, FR, DE, UK
PAT 609	Dosimeter system	US
PAT 559	Electrical power conditioning unit and system	US
PAT 561	Imaging optics and optical device for mapping a curved image field	US
PAT 588	Method and apparatus for generating a digital signal of tunable frequency and frequency synthesizer employing same	US
PAT 572	Method and apparatus for monitoring an operational state of a system on the basis of telemetry data	DE, UK, NL, IT, FR
PAT 554	Method for assembling a mirror plate stack	IT
PAT 566	Method, apparatus and system for determining a position of an object having a global navigation satellite system receiver by processing undifferenced	UK, CN, CH, ES, NL, NO, DE, US
PAT 576	Method, device and system for compressing time series data	DE, UK, ES, IT, FR
GAL 004	Spreading codes for a satellite navigation system	IN

Patents granted in 2016

Events & International Exhibitions

Events and exhibitions either organised by, or with support from, ESA Communications.

VIP visit at ESTEC <i>Noordwijk, Netherlands</i> 6 January	Irish Taoiseach (Prime Minister) Enda Kenny	Citizens' Debate <i>ESA Member States</i> 10 September	About 2000 citizens participate in the first Citizens' Debate on Space for Europe
ESRIN Open Day <i>Frascati, Italy</i> 7–11 March	Annual open day with educational events	European Robotics Congress and Rover Competition <i>Jasionka (Rzeszów), Poland</i> 12–19 September	
Living Planet Symposium <i>Prague, Czech Republic</i> 9–13 May	Largest Earth observation conference ever organised by ESA	New Scientist Live <i>London, UK</i> 22–25 September	ESA stand at exhibition with high national and social media profile
VIP visit to EAC <i>Cologne, Germany</i> 18 May	German Chancellor Angela Merkel	Meeting of Members of Parliament <i>Guadalajara, Mexico</i> 25 September	Theme Connecting the World via Space – Policies, Technologies and Applications', co-organised with IAF
Couture in orbit <i>Science Museum, London, UK</i> 25 May	Fashion show using space materials and technology	ESTEC Open Day <i>Noordwijk, Netherlands</i> 2 October	Large-scale annual open day organised during World Space Week
ILA International Air Show <i>Berlin, Germany</i> 1–4 June	Space pavilion exhibition and a programme of events	European Interparliamentary Space Conference <i>Bucharest, Romania</i> 3–4 October	XVIIIth Plenary Session, ESA Director General as keynote speaker
European Space Solutions Conference <i>The Hague, Netherlands</i> 30 May – 3 June		European Parliament <i>Brussels, Belgium</i> 16 November	Science and Technology Options Assessment lecture organised with ESA, DG gave introductory speech
Farnborough International Airshow <i>Farnborough, UK</i> 11–17 July	Joint exhibition stand for ESA, UK Space Agency and UK space industry	ESA Council at Ministerial Level <i>Lucerne, Switzerland</i> 1–2 December	
Advanced Manufacturing Laboratory <i>Harwell campus, UK</i> 19 July	New laboratory inauguration by ESA Director General Jan Wörner	Royal visit to ESTEC <i>Noordwijk, Netherlands</i> 29 November	King Philippe of the Belgians

Chairs of Council and subordinate bodies 2016

Chair of the Council	Bo Andersen (NO) Jean-Yves Le Gall (FR)	(as of June 2015) (as of June 2015)
Vice-Chair	Klaus Pseiner (AT)	(as of June 2015)
Science Programme Committee	Roel Gathier (NL) † Joergen Christensen-Dalsgaard (DK)	(until March 2016) (from March 2016)
Administrative and Finance Committee	Helene Körling (SE)	
Industrial Policy Committee	Cristina Falvella (IT)	
International Relations Committee	Michal Fridrich (CZ)	
Security Committee	Kai Knape (FI)	
Human Spaceflight, Microgravity and Exploration Programme Board	Pilar Roman Fernandez (ES)	
Launchers Programme Board	Agnès Grandjean (BE)	
Earth Observation Programme Board	Jacques Nijskens (BE)	
Programme Board on Satellite Navigation	Catherine Mealing-Jones (GB)	
Joint Board on Communication Satellite Programmes	Catherine Mealing-Jones (GB)	
Space Situational Awareness Programme Board	Klaus Steinberg (DE)	

National delegations to Council in 2016

AUSTRIA

Klaus Pseiner
Andreas Geisler
Andrea Kleinssasser

BELGIUM

Jacques Nijskens
Frank Monteny

CZECH REPUBLIC

Karel Dobeš
Václav Kobera

DENMARK

Peter Sloth
Gorm Kofoed Petersen

ESTONIA

Madis Võõras
Marika Popp

FINLAND

Ilona Lundström
Kimmo Kanto
Marjaana Aarnikka

FRANCE

Joël Barre
Thierry Duquesne†
Jean-Pascal Le Franc
Donato Giorgi

GERMANY

Pascale Ehrenfreund
Gerd-Achim Gruppe
Karl-Friedrich Nagel

GREECE

Thomas Maloutas
Vassilis Tsoukikas
Mickael Kotsias

HUNGARY

Fruzsina Tari
István Erényi
Szilvia Szántó

IRELAND

Michael Davitt
Tony McDonald

ITALY

Roberto Battiston
Laura Aghilarre
Maria Cristina Falvella

LUXEMBOURG

Marc Serres
Mathias Link

NETHERLANDS

Jasper Wesseling
Frits von Meijenfeldt
Nico van Putten

NORWAY

Geir Hovmork
Knut Myrvang

POLAND

Otylia Trzaskalska-Stroińska
Anna Nałęcz-Kobierzycka

PORTUGAL

Paolo Ferrao
José Carlos Caldeira
Luis Serina

ROMANIA

Marius-Ioan Piso
Oana Maria Neagu

SPAIN

Begoña Cristeto
Francisco Marín
Juan Carlos Cortés Pulido
Jorge Lomba Ferreras

SWEDEN

Olle Norberg
Thorwald Larsson
Maria Nilsson

SWITZERLAND

Raphaël von Roten (until 11 July 2016)
Renato Krpoun (as of 11 July 2016)
Urs Frei

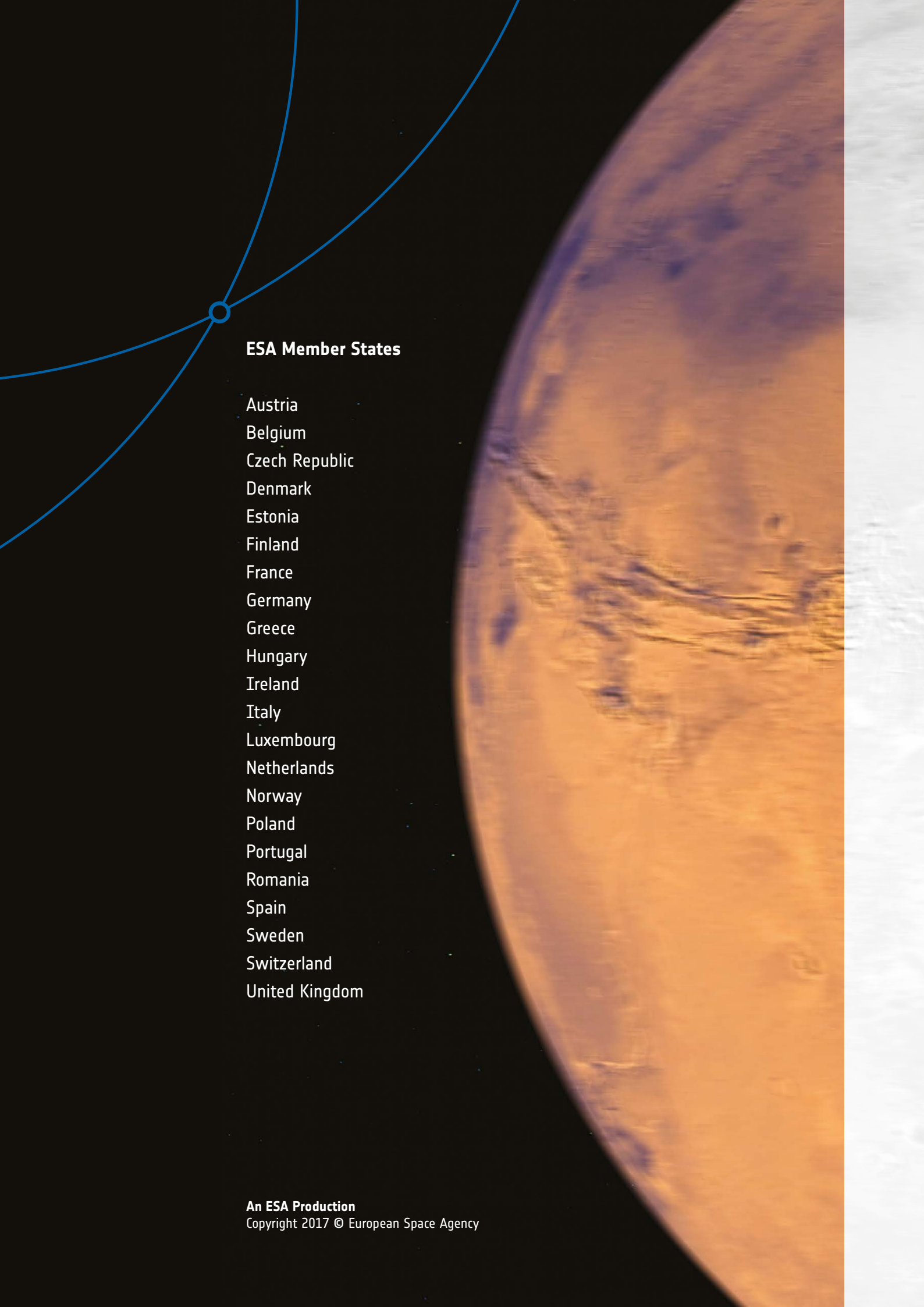
UNITED KINGDOM

Katherine Courtney
Alice Bunn
Mark Bacon

CANADA

Sylvain Laporte
Jean-Claude Piedboeuf
Jean-Pierre Arseneault

[illegible]

The background of the page features a large, detailed image of the planet Mars on the right side, showing its reddish-orange surface with various craters and geological features. On the left side, there is a dark blue background with two thin, curved lines intersecting at a point marked by a small white circle. The text 'ESA Member States' is positioned to the right of this intersection point.

ESA Member States

Austria
Belgium
Czech Republic
Denmark
Estonia
Finland
France
Germany
Greece
Hungary
Ireland
Italy
Luxembourg
Netherlands
Norway
Poland
Portugal
Romania
Spain
Sweden
Switzerland
United Kingdom